INCH-POUND
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PERFORMANCE SPECIFICATION

INVERTERS, AIRCRAFT, GENERAL SPECIFICATION FOR

Reactivated after 10 September 2010 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the general requirements for inverters that convert electrical power in aircraft from nominal 28 volts direct current (VDC) (see 6.5.4) power to three-phase or single-phase alternating current (AC) (see 6.5.2) power at 400 hertz.

1.2 Classification. Inverters are of the following classes, as specified (see 3.5.2.1 and 6.2):

Class A. Class A inverters are self-cooled and designed to meet the temperature-altitude requirements on figure 1, class A.

Class B. Class B inverters are self-cooled and designed to meet the temperature-altitude requirements on figure 1, class B.

Class C. Class C inverters are blast-cooled and designed to meet the temperature-altitude requirements on figure 1, class C.

Comments, suggestions, or questions on this document should be addressed to the Naval Air Systems Command (Commander, Naval Air Warfare Center Aircraft Division, Code 4L8000B120-3, Highway 547, Lakehurst, NJ 08733-5100) or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract. (see 6.2).

FEDERAL STANDARDS

FED-STD-595/37038	Miscellaneous, Flat or Lusterless (Black)

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-15024	Plates, Tags, and Bands for Identification of Equipment, General Specification for

(See supplement 1 for list of specification sheets.)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Electronic and Electrical Component Parts
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	Dissimilar Metals

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AEROSPACE INDUSTRIES OF AMERICA (AIA)

NASM90415 Nut, Self-locking, Steel, 160 KSI, 450°F, 12 Point, Captive Washer (DoD adopted)

(Copies of this document are available from http://www.aia-aerospace.org or Aerospace Industries Association of America, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3928.)

JOINT ELECTRON DEVICE ENGINEERING COUNCIL (JEDEC)

JESD471 Symbol and Label for Electrostatic Sensitive Devices.

(Copies of this document are available from http://www.jedec.org or JEDEC, 3103 North 10th Street, Suite 240-S, Arlington, VA 22201-2107.)

RTCA, INCORPORATED

DO-160 Environmental Conditions and Test Procedures for Airborne Equipment

(Applications for copies should be addressed to the RTCA, Inc., 1828 L Street, NW, Suite 805, Washington, DC 20036-5133, http://www.rtca.org/.)

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE INTERNATIONAL)

SAE-AMS-A-21180 Aluminum-Alloy Castings, High Strength. (DoD adopted)

SAE-AMS-M-3171 Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on. (DoD Adopted)

SAE-AMS-STD-2175 Castings, Classification and Inspection of. (DoD adopted)

SAE-AS7928 Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for. (DoD adopted)

SAE-AS8879 Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter. (DoD adopted)

SAE-AS50881 Wiring Aerospace Vehicle. (DoD adopted)

(Copies of this document are available from www.sae.org or SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern (see 6.13).

3.2 Qualification. Inverters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.3).

3.3 Materials.

3.3.1 Selection of materials, parts, and processes. The materials selected by the contractor shall be capable of meeting all of the operational and environmental requirements specified herein (see 6.6.2 and 6.6.3). Castings shall be in accordance with SAE-AMS-STD-2175, or equivalent. Aluminum alloy castings shall be in accordance with SAE-AMS-A-21180. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnating, welding, or any other method acceptable to the procurement activity. Inspection of repaired castings shall be governed by quality control techniques subject to the approval of the procuring activity.

3.3.2 Moisture resistance. Non-metal materials, including plastics, ceramics, fabrics, and protective finishes, shall be moisture resistant. Non-metal materials and parts may be treated to conform to this requirement. Electrical connectors shall be environment resistant and withstand the environmental tests herein (see 4.5.17 through 4.5.25).

3.3.3 Corrosion resistance. All metals used in the inverter shall be processed and protected to resist corrosion (see 6.6.2.1). Protective coatings and finishes shall provide abrasion resistance as required to withstand the environmental tests specified herein. The use of magnesium is prohibited unless specifically approved for each application by the using service. Where approved, magnesium alloy parts shall be surface treated for protection against corrosion in accordance with SAE-AMS-M-3171. Dissimilar metals shall not contact each other unless protected against electrolytic corrosion in accordance with MIL-STD-889. Corrosion protection applied to dissimilar metals, such as metal plating and coatings, shall withstand the environmental tests herein (see 4.5.17 through 4.5.25).

3.3.4 Effects of fluids. The inverter shall show no deterioration at normal operating temperatures due to wetting from aviation fuel (JP-4/JP-5), fire extinguishing compounds, de-ice fluids, aircraft washing fluids, hydraulic fluids, and lubricating oils (see 4.5.25 and 6.9).

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3.3.5 Toxicity and fire resistance. Materials used shall be flame resistant, shall not support combustion, and shall be non-toxic when exposed to flame when used under all operating and environmental conditions herein (see 4.5.26).

3.3.6 Recycled, recovered, or environmentally preferable materials. The use of toxic chemicals, hazardous substances, and ozone depleting chemicals shall be avoided in the preparation and production of equipment. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 6.6.2.2).

3.3.7 Color. The inverter shall be finished in 37038 black in accordance with FED-STD-595.

3.4 Design and construction. The inverter shall be a single package. Dynamic protective devices (fuses, circuit breakers, and thermally actuated switches) shall not be used. Deviations from this specification or from the applicable specification sheet must be specifically approved by the procuring and qualifying activities (see 4.5.1).

3.4.1 General mechanical and electrical design. Parts and equipment shall provide the performance, reliability, and service life required herein (see 4.5.1). Electronic parts shall be selected to enable the inverter to provide the specified performance, reliability, and service life (see 6.6.3.5). Static discharge control shall be provided for protection of electronic devices during assembly and handling (see 3.5.2.10 and 6.6.3.4).

3.4.2 Electrical.

3.4.2.1 Electric connections.

3.4.2.1.1 External electric connections. The equipment shall be so designed that external connections and wiring can be made in accordance with the wiring installation, ground return, and connector requirements of SAE-AS50881. The equipment structure shall not be used as a current path except for radio noise shielding. External connections necessary for appropriate input and output shall be as shown on the applicable specification sheet.

3.4.2.1.1.1 Terminal block design. Where terminals are used for the external connection to inverters, they shall be of the stud type, and shall be designed so that the current is conducted by means of surface-to-surface contact, and not through the stud threads. All studs shall be steel, corrosion resistant, C-34 Rockwell minimum hardness, and shall be of sufficient length to accommodate two SAE-AS7928 lugs. Nuts used on terminals shall conform to NASM90415. There shall be no dielectric material in the compression build-up. The terminal block shall be so designed that it can be removed and replaced on the inverter without the necessity of rebrazing or soldering. Barriers affording a positive separation of leads and terminals shall be provided on the terminal block. Durable, reusable, non-conductive terminal covers shall be provided. Terminal designations shall be durably, legibly, and prominently marked on the terminal block.

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3.4.2.1.1.2 Connectors. When alternate connectors are specified, the inverter design shall be such that it may be loaded on either receptacle, but not simultaneously. This shall be accomplished by a simple, easily-accessible means to electrically disconnect the unused receptacle, or by providing a protective cover that will automatically cover the unused receptacle. The cover, if used, shall be designed so that it cannot be separated from the inverter.

3.4.2.1.1.3 Output voltage jacks. One set of placarded pin jacks shall be provided on the inverter located as shown on the applicable specification sheet between which the output voltage can be measured. The pin jacks shall accept a probe pin of 0.078 ±0.002 inch in diameter. A set shall consist of two jacks for single-phase inverters and four jacks for three-phase inverters.

3.4.2.1.2 Internal electric connections. Internal connections shall be made in accordance with aerospace industry standards (see 6.6.3.2). For three-phase inverters, the design shall be such that no internal reconnections are required to change from wye to delta.

3.4.2.2 Adjustments.

3.4.2.2.1 Static inverters. Static inverters shall not have any external provisions for voltage or frequency adjustment, nor shall any internal adjustments be necessary at the time of installation. All adjustments shall be made at the time of manufacture or overhaul.

3.4.2.2.2 Rotary inverters. Means for adjusting rotary inverters shall be in accordance with 3.4.2.2.2.1 and 3.4.2.2.2.2.

3.4.2.2.2.1 Voltage setting. When required by the applicable specification sheet, the inverter shall be constructed to permit the output voltage to be set over a range of at least ±5.0 percent of rated value in increments of not more than 0.5 percent of rated voltage. The setting means shall be so constructed that it will not be displaced by vibration or shock. The control shall be designed for adjustment using a straight screwdriver.

3.4.2.2.2.2 Frequency setting. No frequency adjustment capability will be required if the inverter has an internal control that automatically maintains the frequency over the life of the inverter. If a frequency adjustment capability is incorporated in the inverter, the control shall be constructed so that it will not be displaced by vibration or shock, and shall be adjustable with a straight screwdriver over the following ranges:

a. Inverters 250 voltampere (VA) and larger shall have an adjustable frequency range at least ±2.5 percent of the rated frequency in increments of not more than 0.25 percent of the rated frequency.

b. Inverters smaller than 250 VA shall have an adjustable range at least ±5.0 percent of the rated frequency in increments of not more than 0.25 percent of the rated frequency.

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3.4.3 Mechanical.

3.4.3.1 Rotary inverters.

3.4.3.1.1 Rotor. Rotary inverters shall be constructed so that the rotor can be removed without special tools.

3.4.3.1.2 Brushes (brush type inverters only).

3.4.3.1.2.1 Brush accessibility. Brushes shall be accessible for replacement when brush opening covers are removed. Brush opening covers shall be easily accessible and removable without special tools.

3.4.3.1.2.2 Brush fit. The face of each commutator brush shall contact its commutator 100 percent in the direction of rotation for at least 75 percent of brush dimension parallel to the shaft. The face of each slip ring brush shall contact its slip ring over an area that will insure satisfactory performance of the inverter; however, slip ring brush contact area shall be a minimum of 50 percent.

3.4.3.1.2.3 Cartridge type brushes. When cartridge type brush assemblies are used, there shall be an integral assembly of the brush, spring, and terminal which do not rotate during installation or service.

3.4.3.1.2.4 Brush position. Brush riggings for all units of the same part number by one manufacturer shall be completely interchangeable. Brush rigging in each inverter shall be pinned in that position which results in optimum commutation characteristics for that unit. The method of pinning shall not adversely affect interchangeability of parts between units, and shall allow the brush rigging position to be reset during overhaul.

3.4.3.1.2.5 Brush marking. Each brush shall be marked for allowable wear. Brushes having a cross-sectional area parallel to the wearing surface larger than 1/16 inch shall be marked for allowable wear by a readily discernible groove in its edge. Brushes having a cross-sectional area parallel to the wearing surface of less than 1/16 square inch may be marked with a line. In either case, the groove or line shall be as shown on figure 2.

3.4.3.1.2.6 Spare brushes. All spare brushes shall be formed to a radius equal to approximately 100 percent of the radius of the new commutator or slip ring, and shall be of the same specifications as the corresponding brushes approved by the qualification tests.

3.4.3.1.2.7 Brush life. Brushes shall have a useful life of not less than 500 operating hours.

3.4.3.1.3 Lubrication. Bearings and other lubricated parts shall not require additional lubrication during 1,000 hours of operation.

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3.4.3.2 Static inverters. Static inverters shall not have any moving parts except that a cooling fan shall be allowable.

3.4.3.3 Interchangeability. Design tolerances shall permit parts, subassemblies, and assemblies to be used in the inverter without regard to source of supply or manufacturer. Parts, subassemblies, and assemblies having the full range of dimensions and characteristics permitted by governing specifications shall be usable as replacement items without selection and without departure from the specified performance requirements of the inverter.

3.4.3.4 Mechanical connections, fasteners, inserts, and hardware retention. Bolts and screws shall be in standard size and thread forms selected in accordance with aerospace industry standards (see 6.6.2.3). Inch-pound standard type screws shall conform to the UNJ thread form series in accordance with SAE-AS8879. All threaded parts, internal and external to the equipment, shall be positively locked. Self-locking nuts, safety wiring, or acceptable alternative methods shall be used to prevent accidental loosening of threaded parts. Staking shall not be used. Internal devices with provisions for mechanical adjustments shall be locked and sealed during production.

3.4.3.5 Cooling. The inverter shall be self-cooled. No conduction cooling to the inverter environment shall be allowed. The stabilized operating temperature of the inverter, while operating over the range of environmental and electrical loading conditions specified herein, shall prevent the individual inverter components from exceeding their rated temperatures (see 6.6.2). The use of an internally mounted, integral cooling fan is permitted for inverters with ratings of 750 VA and greater (see 6.6.2.5). For fan-cooled inverters the direction of air flow shall be as specified in the applicable specification sheet.

3.4.3.6 Ventilation openings. All ventilation or other openings shall be designed to prevent passage of foreign objects, and shall be of such size to prevent passage of spherical objects of 5/16 inch diameter or larger.

3.4.3.7 Operating position. The inverter shall operate in any position.

3.4.3.8 Size and weight. The dimensions and weight shall be as specified in the applicable specification sheet. The weight and dimensions shown on the applicable specification sheet shall include the weights and dimensions of all auxiliary apparatus necessary to make the inverter conform to this specification and the applicable specification sheet. Where only maximum and minimum dimensions are shown, the part need not have the shape shown; however, the part, including all protrusions, shall be contained within the outline shown (see 4.5.1).

3.5 Performance characteristics. The inverter shall deliver power in accordance with the requirements specified herein and the applicable specification sheet. For three-phase inverters, the limits apply to each phase.

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3.5.1 Electrical.

3.5.1.1 Input power. The inverter shall meet the requirements of this specification when supplied with power as defined herein from an aircraft electrical system. The nominal input voltage is 28 volts DC.

a. Normal input voltage (all inverters) - The normal input voltage range (see 6.5.8) is indicated on the applicable specification sheet and is typically 22 to 29 volts DC.

b. Abnormal input voltage (all inverters) - The abnormal input voltage range (see 6.5.1) is indicated on the applicable specification sheet and is typically 20 to 31.5 volts DC.

c. Low input voltage - The low input voltage range (see 6.5.7), when applicable, is indicated on the applicable specification sheet and is typically 12 to 20 volts DC.

3.5.1.2 Output voltage regulation. The inverter shall have a nominal output voltage of 115 VRMS and maintain the following steady state output voltages under all specified conditions (see 4.5.2 and 4.5.4):

a. Normal input voltage (see 3.5.1.1a)

(1) Single-phase and balanced three-phase loads: 112.5 - 117.5 volts RMS

(2) One-third unbalanced three-phase loads: 112.5 - 118.0 volts RMS

(3) Two-thirds unbalanced three-phase loads: 112.0 - 119.0 volts RMS

b. Abnormal input voltage (see 3.5.1.1b) for single-phase and balanced three-phase loads: 108.0 - 117.5 volts RMS

c. Low input voltage (inverters required to supply regulated output at low input voltages) (see 3.5.1.1c) for single-phase and balanced three-phase loads: 108.0 - 117.5 volts RMS

3.5.1.3 Output frequency regulation. The inverter shall have a nominal output frequency of 400 hertz. Rotary inverters shall maintain the steady-state frequency as indicated on the applicable specification sheet, and static inverters shall maintain a steady-state frequency of 400 ±7 hertz under all specified input voltages (see 3.5.1.1), load, temperature, and altitude conditions specified (see 4.5.2).

3.5.1.4 Efficiency. The efficiency of rotary inverters shall be as specified in the applicable specification sheet. The efficiency of static inverters shall be as specified in table I (see 4.5.3).

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3.5.1.5 Three-phase output.

3.5.1.5.1 Phase rotation. The phase rotation shall be positive, in the order of A-B-C, at angles of 0°, 120°, and 240° (see 4.5.4.2).

3.5.1.5.2 Output voltage unbalance. For all balanced and one-third unbalanced load conditions, the maximum output voltage unbalance (see 6.5.9) measured between any two phases shall be 3 volts RMS. For two-thirds unbalanced load conditions, the maximum output voltage unbalance measured between any two phases shall be 4 volts RMS (see 4.5.4.1.1). Individual phase voltages (see 6.5.10) shall be within the limits specified in 3.5.1.2.

3.5.1.5.3 Phase displacement. The displacement between adjacent phases shall be 120 ± 4° for both balanced and unbalanced load conditions (see 4.5.4.3).

3.5.1.6 Recovery time. When rated full load (see 6.5.5) is applied and removed, the output voltage shall remain within the limits of figure 3 (see 4.5.5).

3.5.1.7 Input power application.

3.5.1.7.1 Static inverters. The output voltage during input power application shall be within and remain within the limits of the applicable specification sheet within two seconds after application of input voltage. During the two-second period, output voltage transients shall be within the limits of figure 3. For determining compliance of voltage transients to figure 3, the onset of the transient shall be considered as the time when the output voltage first reaches a value equal to minimum steady state limits (see 3.5.1.2). The input power application shall not exceed 1000 percent of full load current as computed from the minimum allowable efficiency specified (see 3.5.1.4). The current shall return to and stay within 150 percent of full load value within 0.1 second after first application of input voltage (see 4.5.6.1 and 4.5.6.2).

3.5.1.7.2 Rotary inverters. The uncontrolled output voltage during input power application shall not exceed 105 percent of the maximum controlled valued specified in the applicable specification sheet after 0.25 second. During the 25-second warm-up period of figure 4, the output voltage shall not exceed the maximum steady-state voltage as specified in the applicable specification sheet, except that one transient excursion of up to two seconds is permitted, not to exceed 140 VRMS. The onset of this transient shall be considered as the time when the output voltage first exceeds the maximum steady-state voltage as specified in the applicable specification sheet. The input power application shall not exceed 1000 percent of full load current as computed from the minimum allowable efficiency specified on the applicable specification sheet or 1,500 amperes, whichever is less. The current shall return to and stay within 300 percent of full load value within 0.3 second after first application of input voltage (see 4.5.6.1 and 4.5.6.2).

3.5.1.8 Overload and short circuit capacity. Over the normal input voltage range (see 3.5.1.1a) the inverter shall be capable of operating without damage into any passive load impedance (including a short circuit) for five seconds. For three-phase inverters, this shall apply to each phase separately, any two phases simultaneously, or all phases simultaneously. The

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inverter shall be capable of meeting all the overload and short circuit requirements with two minutes separating the removal and subsequent application of an overload or short circuit (see 4.5.7).

3.5.1.8.1 Overload. Over the normal input voltage range (see 3.5.1.1a), inverters rated 1000 VA or less shall, without damage, maintain the minimum output voltage specified by the applicable specification sheet while delivering 200 percent rated output current for five seconds. Inverters rated above 1000 VA shall maintain the minimum output voltage specified by the applicable specification sheet while delivering 150 percent rated output current for 15 seconds. For three-phase inverters, this shall apply to each phase separately, any two phases simultaneously, or all phases simultaneously. During these conditions, no phase voltage shall exceed 117.5 volts RMS.

3.5.1.8.2 Short circuit. Over the normal input voltage range (see 3.5.1.1a), inverters rated 1000 VA or less shall be capable of delivering, without damage, at least 250 percent rated load current for five seconds. Inverters rated above 1000 VA shall be capable of delivering, without damage, at least 200 percent rated load current for five seconds. For three-phase inverters, short circuit requirements shall apply to one phase, two phase, and three phase line-to-line and line-to-neutral short circuits. During these conditions, no phase voltage shall exceed 117.5 volts RMS.

3.5.1.8.3 Protection (static inverters only). After delivering current into any overload or short circuit condition for at least five or 15 seconds (as applicable), the inverter shall have automatic electronic overload and short circuit protection and shall then operate continuously for at least 15 minutes, without damage, while connected to the overload or short circuit. The output voltage and/or current may drop to any level after the initial five or 15 seconds (as applicable). Normal output voltage shall be automatically restored within two seconds after the removal of the overload or short circuit. For three-phase inverters this requirement shall apply to each phase separately, any two phases simultaneously, or all phases simultaneously.

3.5.1.9 Electromagnetic compatibility. The inverter shall conform to the requirements of MIL-STD-461 when tested to the requirements as specified in 4.5.8. The inverter shall not interfere with the performance of the other equipment as a result of radiated or conducted electromagnetic interference. In addition, the inverter performance shall not be degraded as a result of electromagnetic interference from other aircraft equipment.

3.5.1.10 Output modulation (see 4.5.9).

3.5.1.10.1 Voltage. The output voltage modulation shall not exceed 1.0 percent under normal input voltage conditions at full load, half load, and no load, and 1.5 percent under abnormal and low input voltage conditions (see 3.5.1.1 and 4.5.9.1).

3.5.1.10.2 Frequency. The inverter shall not develop sustained frequency oscillations under conditions of full load, half load, or no load. For static inverters the output frequency modulation shall not exceed 0.7 percent, nor shall the deviation rate exceed 200 hertz per second (see 4.5.9.2).

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3.5.1.11 Waveform.

3.5.1.11.1 Linear loads (full wave sinusoidal output current). The output voltage crest factor (see 6.5.3) shall not exceed the limits of 1.414 ±0.1 over the normal input voltage range. The crest factor shall not exceed the limits of 1.414 ±0.3 over the abnormal input voltage range. For static inverters the total harmonic and extraneous frequency content of the waveform shall not exceed 5 percent of the fundamental, nor shall any individual harmonic or extraneous frequency exceed 3 percent of the fundamental, over the normal input voltage range (see 3.5.1.1). At temperature and altitude conditions outside of those specified in 4.4.1.1 and 4.4.1.2 the 3 percent limit shall not apply (see 4.5.10.1.1).

3.5.1.11.2 Nonlinear loads (static inverters only). During the tests specified in 4.5.10.1.2, the total harmonic and extraneous frequency content of the voltage waveform shall not exceed 5 percent of the fundamental.

3.5.1.12 Input ripple current (static inverters only). Peak-to-mean input ripple currents shall not exceed the values specified in table II when the inverter is operated from a low impedance, low ripple source (see 4.5.11).

3.5.1.13 Input power susceptibility (static inverters only). Input surge and spike voltages defined in 4.5.12 shall not damage the inverter. There shall be no evidence of flashover or insulation breakdown as a result of the tests specified in 4.5.12 and the output transients generated during the test shall not exceed the limits defined on figure 3. During the test specified in 4.5.12.3, the output voltage, frequency, output modulation, and waveform of the inverter shall be within the limits specified in 3.5.1.2a, 3.5.1.3, 3.5.1.10, and 3.5.1.11.

3.5.1.14 Extreme input voltage (static inverters only). The inverter shall be capable of carrying rated load when operated continuously at the minimum abnormal input voltage (see 3.5.1.1.b). The inverter shall not be damaged when energized continuously at input voltages from zero to 32 volts under any rated load condition. The inverter shall maintain output voltage between the maximum normal input, and 32 volts input, however, no other requirements shall apply except output voltage shall not exceed 117.5 volts RMS under these conditions (see 4.5.13).

3.5.1.15 Reduced input voltage operation (rotary inverters only). During the reduced input voltage test (see 4.5.14) the output voltage of inverters rated at 750 VA and below shall remain between 100 and 120 VRMS, and the frequency between 350 and 420 hertz. For inverters rated above 750 VA the output voltage shall remain between 100 and 120 volts RMS, and the frequency between 370 and 410 hertz.

3.5.1.16 Acoustical noise. When the applicable specification sheet specifies limits and when tested as specified in 4.5.15, the maximum sound level measured at any point shall not exceed the specified limits.

3.5.1.17 Overspeed (rotary inverters only). The inverter shall not fail when subjected to the test specified in 4.5.16.

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3.5.2 Environmental.

3.5.2.1 Temperature-altitude. The inverter shall be capable of operation without degradation of performance throughout the temperature-altitude range specified below for the class specified in the applicable specification sheet (see 4.5.2.3 and 4.5.2.4).

3.5.2.1.1 Class A. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range on figure 1, class A (see tables V and VI).

3.5.2.1.2 Class B. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range on figure 1, class B (see tables V and VI).

3.5.2.1.3 Class C. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range on figure 1, class C (see tables V and VI).

3.5.2.2 Acceleration. The output voltage and frequency of the inverter shall remain within the limits of 3.5.1.2 and 3.5.1.3 while operating during the acceleration test (see 4.5.17). There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. The voltage and frequency regulation after the test shall be within the limits specified in 3.5.1.2 and 3.5.1.3.

3.5.2.3 Humidity. As a result of the humidity test there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.5.1.2 and 3.5.1.3 (see 4.5.18).

3.5.2.4 Shock (see 4.5.19).

3.5.2.4.1 Functional. The output voltage shall not exceed the limits of figure 3 while operating during the functional shock test. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.5.1.2 and 3.5.1.3 (see 4.5.19.1).

3.5.2.4.2 Crash hazard. The inverter shall remain in place and not create a safety hazard during the crash hazard shock test (see 4.5.19.2).

3.5.2.5 Vibration. The inverter shall withstand the vibration environment specified herein (see 4.5.20 and 6.8). The output voltage modulation of the inverter measured during vibration shall not exceed two percent. The output voltage and frequency measured during vibration shall remain within the limits specified in 3.5.1.2 and 3.5.1.3. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion

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of the test, the voltage and frequency regulation shall be within the limits specified in 3.5.1.2 and 3.5.1.3.

3.5.2.6 Fungus. The inverter shall be constructed of non-nutrient materials that do not support the growth of fungus in aircraft operating environments (see 4.5.21).

3.5.2.7 Salt fog. As a result of the salt fog test there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.5.1.2 and 3.5.1.3 (see 4.5.22).

3.5.2.8 Dust. The inverter shall withstand the effects of dust from world-wide operation, as simulated by the dry dust test of 4.5.23.

3.5.2.9 Explosive atmosphere. The inverter and connections shall not cause ignition to any surrounding explosive mixture when tested as specified in 4.5.24. Seals and coatings providing ignition hazard protection shall be undamaged by the tests in section 4.5.

3.5.2.10 Electrostatic discharge (ESD). The inverter shall be unharmed by electrostatic discharge through contact with personnel.

3.6 Reliability. Inverters shall have a minimum observed Mean Flight Hours Between Failure (MFHBF) of 2,000 hours when operated under conditions specified herein. The inverter shall have a minimum time to first maintenance action of 500 flight hours. When used, brushes may be replaced after 500 hours of operation.

3.6.1 Reliability development. The reliability of the inverter shall be demonstrated by a test, analyze, and fix (TAAF) test program (see 4.6).

3.7 Markings.

3.7.1 Terminals. Markings of external electrical connections shall be as specified on the applicable specification sheet (see 4.5.1).

3.7.2 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130 (see 4.5.1and 6.2).

3.7.3 Nameplate. A nameplate conforming to MIL-DTL-15024, type A, B, C, G, or H, with the following markings, in addition to the requirements of MIL-STD-130, shall be securely attached to the inverter. The nameplate shall contain the following information (see 4.5.1):

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Inverter _____ VA
Input: _____ volts DC _____ Amperes DC
Output: _____ volts AC _____ Amperes AC
_400_ Hz _____ Phase

Military Part No.
National Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Date of manufacture
Contract or Order No.
Manufacturer's name or trademark
Inspector's stamp
US

3.7.4 Airflow. When an integral cooling fan is used, direction of airflow shall be plainly and permanently marked on the outside of the housing in the direction specified in the specification sheet. When the direction of the airflow is marked on the cover, provision shall be made to assure that the cover cannot be installed to indicate a false airflow direction.

3.7.5 Rotation marking (rotary inverters only). The direction of rotation shall be plainly indicated by a permanent marking in the form of a curved arrow (see 4.5.1).

3.7.6 Output voltage jacks. Jack designations shall be durably, legibly, and prominently marked on the inverter (see 3.4.2.1.1.3 and 4.5.1).

3.7.7 ESD marking. ESD sensitive assemblies and equipment shall be marked as follows:

3.7.7.1 ESD sensitive assemblies. ESD sensitive assemblies shall be marked with the JESD471 symbols as illustrated below. The symbols shall be located in a position readily visible to personnel when the assembly is incorporated in its next higher assembly.



3.7.7.2 ESD sensitive equipment. Inverters containing ESD sensitive parts and assemblies shall be marked with the JESD471 symbols. The symbols shall be located on the exterior surface of the inverter and readily visible to personnel prior to gaining access to ESD sensitive parts and assemblies within the inverter. The ESD caution statement shown below shall be placed adjacent to the ESD sensitive symbol.

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CAUTION:

CONTAINS PARTS AND ASSEMBLIES
SUSCEPTIBLE TO DAMAGE BY
ELECTROSTATIC DISCHARGE

3.8 Workmanship. The inverter shall be free from irregularities or defects that could degrade performance or durability (see 4.5.1 and 6.6).

3.8.1 Burn-in (static inverters only). Each inverter shall be subjected to a burn-in test as specified in 4.8 in order to expose and eliminate workmanship and component quality problems.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection.

4.2.1 Qualification sample submittal. The qualifying activity will acquire inverters for qualification or unsolicited samples may be submitted to the qualifying activity by manufacturer's seeking qualification (see 6.2 and 6.3). The typical qualification requires four inverters (qualification with less than four samples requires qualifying activity approval). The test schedule matrix in table III is based on four test samples for inverter qualification.

4.2.2 Qualification inspections and tests. Qualification inspection shall consist of the examinations and tests required by this specification and the applicable specification sheet. The manufacturer or an independent laboratory may conduct qualification testing with approval from the Government qualifying activity.

a. Laboratory qualification test - The examinations and tests listed in table III shall be conducted on the inverters submitted for qualification. In addition, the qualifying activity shall have the following options:

(1) the qualification test matrix listed in table III is subject to alteration such that any test may be conducted on any sample submitted for qualification;

(2) inspections may be conducted to verify conformance to any of the requirements that were qualified by similarity;

(3) additional tests/repeated tests may be conducted to determine specification compliance when the measured performance is at specification limits and evidence of

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compliance/non-compliance is inconclusive due to instrumentation accuracy/measurement tolerance.

4.2.3 Qualification retention. Inspection requirements for qualification retention samples shall consist of the examinations and tests listed in table III in accordance with the specification and applicable specification sheet. Samples shall be randomly selected from a production run and submitted to the qualifying activity. Products may be removed from the Qualified Products List if: (1) the manufacturer has not provided the requested certification; (2) the manufacturer has requested that the product be removed from the list; or (3) the qualifying activity has removed the product pending the results of re-qualification testing.

a. The qualifying activity will conduct an evaluation to determine if re-qualification test is necessary when any of the following conditions occur:

(1) change in manufacturer's product design, materials, or processes;

(2) change in performance levels or functional requirements due to specification revision;

(3) production facility relocation; or

(4) company sale or transfer to new ownership.

b. Components demonstrating satisfactory performance to tests conducted for qualification retention will be approved in writing by the qualifying activity for retention on the applicable qualified products list.

4.2.4 Rejection, resubmittal, and retest. Requirements for resubmittal or retest of repaired or reworked test samples will be covered in the contract or purchase order.

4.3 Conformance inspection. Conduct conformance inspection tests in accordance with table IV and the applicable specification sheet. The qualifying activity may conduct conformance inspection on any production components. Unless otherwise directed by the Government contracting officer, delivery of production components should not be delayed pending:

(1) results of any test inspection; or

(2) correction of failures/defects disclosed during conformance inspection. Production lot samples will be selected at random by a Government inspector from production lots and forwarded to the qualifying activity under the contract provisions. Production samples forwarded to the qualifying activity will be subjected to conformance verification inspection in accordance with this specification. At the conclusion of testing, the Government will forward a report of test results to the manufacturer.

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4.4 Standard test conditions.

4.4.1 Test conditions. Unless otherwise specified herein or in MIL-STD-810 or the specification sheet, tests shall be conducted under the following conditions.

4.4.1.1 Ambient temperature. Tests shall be performed at an ambient temperature of 25 ±10°C (77 ±18°F).

4.4.1.2 Ambient altitude. Tests shall be run at approximately sea level altitude.

4.4.1.3 Tolerances for controlled test environment. Unless otherwise specified by the tests of 4.5, the limits for test conditions shall be as follows:

a. Chamber Temperature: ±2 °C (±3.6 °F). (Exclusive of instrument accuracy.)

b. Altitude: ±5 percent.

c. Relative Humidity: ±5 percent

d. Airflow: ± of specified value

e. Vibration Amplitude: Sinusoidal: ±10 percent

f. Vibration Frequency: ±2 percent, or ±1/2 Hz below 25 Hz.

g. Acceleration: ±10 percent.

4.4.1.4 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above, the inverter, and through averaging the results. For inverters without fans, air temperature measurements shall be taken 1 ±1/4 inch from the surface of the inverter. For inverters with an internal fan, air temperature measurements taken 1 ±1/4 inch from the fan inlet shall be considered the ambient temperature.

4.4.1.5 Airflow. For all maximum temperature-altitude tests below 20,000 feet conducted on inverters without integral cooling fans, the airflow velocity over the inverter within the test chamber shall be adjusted to approximate natural convection conditions and shall not exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance. For tests above 20,000 feet altitude, the airflow may exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance in 4.4.1.1.

4.4.1.6 Warm-up and stabilization. Prior to testing, the inverter shall be operated, delivering full rated load specified in the applicable specification sheet for sufficient time for the inverter's temperature, output voltage and frequency to stabilize (see 6.5.5). The inverter shall be

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considered to have stabilized when the change over a ten (10) minute period is measured to be less than:

a. Output voltage: 0.04 volts RMS

b. Frequency: 0.1 Hz

c. Inverter critical component temperature: 0.5 °C (0.9 °F)

4.4.1.7 Voltage measurement. The input and output voltage shall be measured at the terminals of the inverter.

4.4.1.8 Position. The inverter shall be mounted base down.

4.4.1.9 Loading. All tests shall be performed with the inverter operating at full rated load (see 6.6.5).

4.4.1.9.1 Load power factor. Unity power factor load shall be applied.

4.4.1.9.2 Multiple phase loading. For three-phase tests, the loads on each of the three phases shall be balanced with each phase current within 3 percent of the average magnitude of all three phase currents.

4.4.1.10 Input voltage. Nominal rated input voltage shall be applied (see 3.5.1.1).

4.5 Methods of inspection.

4.5.1 Visual and mechanical inspection. The inverter shall be inspected to determine compliance with the requirements of this specification and the applicable specification sheet with respect to materials, design and construction, parts, workmanship, dimensions, weight, and markings for which there are no applicable tests (see 3.3.1, 3.4, 3.4.3.8, 3.7.1, 3.7.2, 3.7.3, 3.7.5, and 3.8).

4.5.2 Voltage and frequency regulation. The output voltage and frequency shall be measured under the conditions specified in 4.5.2.1 through 4.5.2.6. For each load, input voltage, position, and temperature-altitude condition, output voltage and frequency shall remain within the limits specified (see 3.5.1.2 and 3.5.1.3). During the course of the tests, no components of the inverter shall exceed their maximum rated temperatures (see 6.6.2). Tests may be conducted at any intermediate input voltage, load, or temperature-altitude condition within the range on figure 1, class A, B, or C (as applicable) at the option of the activity responsible for qualification.

4.5.2.1 Voltage and frequency regulation at room ambient temperature-altitude conditions. The following tests, (a) through (d) below, shall be conducted immediately upon energizing the inverter, and then repeated after operating the inverter at nominal input voltage, full load, unity power factor until the output voltage and frequency are completely stabilized (see 4.4.1.6).

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a. At the maximum normal input voltage (see 3.5.1.1.a), the load shall be varied from no load to full rated load in steps of one-fourth rating. At each load condition, the inverter shall demonstrate its ability to activate.

b. At nominal rated input voltage (see 3.5.1.1), the test specified in 4.5.2.1.a shall be repeated.

c. At the minimum normal input voltage (see 3.5.1.1.a), the test specified in 4.5.2.1.a shall be repeated.

d. At the minimum abnormal input voltage (see 3.5.1.1.b), the inverter output load current shall be varied from no load to full rated load current (computed on the basis of 115 volts RMS output) in one-fourth load increments. At each load condition, the inverter shall demonstrate its ability to activate.

4.5.2.2 Power factors. The tests specified in 4.5.2.1 shall be repeated at the minimum lagging and minimum leading power factors specified in the applicable specification sheet. Tests conducted at any other load power factors between minimum lagging and minimum leading are at the option of the activity responsible for qualification.

4.5.2.3 Voltage and frequency regulation at maximum temperature-altitude conditions. The inverter shall be tested under each of the temperature altitude conditions listed in table V. Prior to being energized, the inverter shall be soaked at the initial maximum temperature, sea level condition for at least one-half hour. Immediately after energizing the inverter, the test specified in 4.5.2.1 shall be conducted. The tests of 4.5.2.1 and 4.5.2.2 shall then be conducted after operating the inverter at nominal input voltage, full load, unity power factor until inverter output voltage, frequency, and temperature are completely stabilized. For each subsequent temperature-altitude condition, the test of 4.5.2.1 need only be conducted after inverter stabilization.

4.5.2.4 Voltage and frequency regulation at minimum temperature-altitude conditions. The tests specified in 4.5.2.3 shall be repeated substituting the temperature-altitude conditions of table VI and increasing the soak time at the initial -55°C sea level condition to two hours.

4.5.2.5 Operating position. The inverter shall be operated at nominal input voltage, full load, and unity power factor in the normal base down position until output voltage, frequency, and critical component temperatures stabilize. Output voltage, frequency, and critical component temperatures shall be measured as the inverter is rotated by increments of 90 degrees from the normal base down position, 360 degrees about each major axis except the vertical axis. Each position shall be held until output voltage, frequency, and critical component temperatures stabilize.

4.5.2.6 Worst condition. The inverter shall be operated at the worst indicated conditions determined from the tests specified in 4.5.2.1 through 4.5.2.5 over the normal input voltage range. The conditions from each of these tests resulting in highest output voltage shall be simultaneously applied. Prior to the test, the inverter shall be soaked at the applicable ambient temperature for a

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minimum of two hours prior to energizing the inverter. The inverter shall be operated until voltage, frequency, and critical component temperatures stabilize. The above test shall be repeated for minimum voltage, maximum frequency, minimum frequency, and maximum critical component temperature conditions.

4.5.3 Efficiency. The power output and input of the unit shall be measured at full load, unity power factor, sea level, and nominal input voltage. The efficiency shall be calculated under these conditions and shall not be less than specified in 3.5.1.4 under stabilized test conditions as required in 4.4.1.6. When computing efficiency, mean (average) values of input voltage and current shall be used, and root-mean-square (RMS) values of the output voltage and current shall be used.

4.5.4 Three-phase output (three-phase inverters only).

4.5.4.1 Output voltage unbalance (see 3.5.1.2 and 6.5.9).

4.5.4.1.1 Unbalanced loads. The effects of single-phase and unbalanced three-phase loads on the balance of the three-phase voltage output of the unit shall be determined as follows at minimum, nominal, and maximum normal input voltage.

a. When the unit is carrying no three-phase load, a single-phase line-to-neutral unity power factor load requiring 1/3 and then 2/3 of full rated load output current for a single phase (a load of approximately 1/9 and 2/9 respectively of the full three-phase rating of the unit) shall be connected successively to each phase.

b. When the unit is carrying a unity power factor balanced three-phase load requiring 1/3 of the rated full load output current, a single-phase line-to-neutral unity power factor load requiring 1/3 and then 2/3 of the rated full load output current for a single phase (a load of approximately 1/9 and 2/9 respectively of the full three-phase rating of the unit) shall be connected successively to each phase.

c. When the unit is carrying a unity power factor balanced three-phase load requiring 2/3 of the rated full load output current, a single-phase line-to-neutral unity power factor load requiring 1/3 of the rated full load output current for a single phase (a load of approximately 1/9 of the full three-phase rating of the unit) shall be connected successively to each phase.

The maximum value of the output voltage unbalance (see 6.5.9) and the phase voltage occurring during parts (a) through (c) above shall not exceed the limits specified in 3.5.1.5.2.

4.5.4.1.2 Balanced loads. For all balanced loads within the power factor and load rating of the unit, the three-phase output voltage shall comply with 3.5.1.2 and 3.5.1.5.2. This shall be demonstrated during the test specified in 4.5.2.

4.5.4.2 Phase rotation. Compliance of the phase rotation of the inverter with 3.5.1.5.1 shall be determined and verified as part of the test specified in 4.5.4.1.

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4.5.4.3 Phase displacement angle. Compliance of the phase displacement angle of inverter with 3.5.1.5.3 shall be determined and verified as part of the test specified 4.5.2.1, 4.5.2.2, and 4.5.4.1.2.

4.5.5 Recovery time. The output voltage transient shall be measured when rated full load is applied and removed at unity and minimum lagging power factors. The unit shall not develop sustained oscillations and the output voltage shall remain within the limits specified in 3.5.1.6. This test shall be conducted at minimum, ambient, and maximum temperatures specified at approximately sea level pressure and minimum, nominal, and maximum normal input voltages (see 3.5.1.1a).

4.5.6 Input power application (see 3.5.1.7).

4.5.6.1 Minimum voltage. The inverter shall demonstrate the ability to activate and operate continuously with the minimum abnormal input voltage applied at zero, half, and full load (see 3.5.1.1b).

4.5.6.2 Input power application current and voltage. The input power application current and output voltage shall be measured at minimum, nominal, and maximum normal input voltage (see 3.5.1.1.a) while initially applying power to the unit at room ambient temperature at zero, half, and full rated load at unity and minimum lagging power factors. The input power application current and voltage measurements shall indicate compliance with the limits specified in 3.5.1.7. This test shall be conducted at minimum, maximum, and room ambient temperatures specified in 4.4.1.1, table V, and table VI at approximately sea level pressure.

4.5.6.2.1 Cyclic operation (rotary inverters only). The inverter shall be subjected to 50 consecutive five-minute cycles of one-minute operation, and four minutes inoperative. The voltage and frequency shall be within the limits specified in 3.5.1.2 and 3.5.1.3 within one second after each energizing operation, except the first. Upon completion of this test, the inverter shall pass the tests specified in 4.5.2.1a, b, and c. There shall not be any excessive burning nor wear of the brushes.

4.5.7 Overload and short circuit capacity. The inverter shall be tested to demonstrate compliance with the overload and short circuit capacity requirements specified in 3.5.1.8. Overload tests shall be conducted at both unity and minimum lagging power factors. The five second (or 15-second as applicable) overload or short circuit test shall be conducted at maximum, minimum, and ambient temperatures specified at approximately sea level pressure (see 3.5.1.8, table V, and table VI). Compliance to the 15-minute duration overload or short circuit requirement of 3.5.1.8.3 shall be demonstrated only at minimum and room ambient temperatures. For brush-type inverters only, immediately following the runs of 3.5.1.8.1 and 3.5.1.8.2, with the inverter hot, the commutation of all brushes shall be observed over the normal input voltage range for rated load, half load, and no load. There shall be no more than fine pinpoint sparking during this test.

4.5.8 Electromagnetic compatibility. The inverter shall be tested and be in accordance with the requirements and applicable limits of MIL-STD-461. All inverters must meet the requirements of CE102, CS101, CS114, CS115, CS116, RE102, and RS103. All output leads,

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including the neutral, must comply with the requirements of CE102. Inverters used in Army or Anti-submarine Warfare (ASW) applications must also meet the requirements of CE101, RE101, and RS101. The test shall be conducted at full load and half load, at minimum and maximum normal input voltage. The inverter load power factor shall be adjusted such that the combined power factor of the load and all auxiliary test equipment required shall present a total load of unity power factor to the inverter. During electromagnetic susceptibility testing, the output voltage and frequency shall remain within the normal limits of 3.5.1.2 and 3.5.1.3. All testing shall be conducted by National Association of Radio and Telecommunications Engineers (NARTE) certified personnel at a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

4.5.9 Output modulation. The inverter shall comply with the requirements of 3.5.1.10.1 and 3.5.1.10.2 when operated at the conditions specified in 4.5.2.1 through 4.5.2.4, except that frequency modulation need only be measured during the tests of 4.5.2.1 and 4.5.2.2.

4.5.9.1 Voltage modulation. Total output voltage modulation shall be determined by means of a calibrated oscilloscope or other means satisfactory to the responsible qualifying activity. The percent modulation shall be computed as follows:

$$\text{Percent Modulation} = \frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100\%$$

E_{max} and E_{min} are the maximum and minimum peak voltages respectively, measured for the same polarity during a one-second time period.

4.5.9.2 Frequency modulation. Output frequency modulation shall be determined by a method acceptable to the responsible qualifying activity. The percent modulation shall be computed as follows:

$$\text{Percent Modulation} = \frac{F_{max} - F_{min}}{F_{max} + F_{min}} \times 100\%$$

F_{max} and F_{min} are the maximum and minimum instantaneous frequencies, respectively, attained during a 15-second time period.

4.5.10 Waveform.

4.5.10.1 Distortion. The harmonic and extraneous frequency content of the output voltage, per phase for three-phase inverters and across the two output lines for single-phase inverters shall not exceed the limits specified in 3.5.1.11.1 and 3.5.1.11.2 when tested in accordance with 4.5.10.1.1 and 4.5.10.1.2.

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4.5.10.1.1 Linear loads (full wave sinusoidal output current).

a. Under the test conditions specified in 4.5.2.1a, b, and c with unity power factor, steady-state load.

b. Under the test conditions specified in 4.5.2.1a, b, and c with steady-state minimum lagging power factor load specified in the applicable specification sheet.

c. Tests (a) and (b) above shall be repeated at maximum and minimum rated temperatures at approximately sea-level pressure.

4.5.10.1.2 Non-linear loads (static inverters only). The tests specified below shall be conducted at minimum, nominal, and maximum normal input voltages (see 3.5.1.1a). Tests on three-phase inverters may be conducted one phase at a time.

a. With an initial linear load of 50 percent rated current, an additional load of 50 percent rated current shall be added for approximately 1/2 of each positive and negative half cycle.

b. Part (a) shall be repeated except that the load during the positive half cycle shall be linear and 100 percent rated current.

c. With no initial load, a load of 50 percent rated current shall be applied for approximately 1/2 of each positive and negative half cycle.

d. Part (c) shall be repeated except that the load during the positive half shall be linear and 50 percent rated current.

4.5.10.2 Crest factor. The crest factor (see 6.5.3) shall be measured under the test conditions specified in 4.5.2.1 and 4.5.2.2 to verify compliance with 3.5.1.11.1. The crest factor limits apply to either polarity separately.

4.5.11 Input ripple current (static inverters only). The input ripple current shall be measured with an oscilloscope under the test conditions specified in 4.5.2.1a, b, and c, using a low impedance, low ripple source of power to verify compliance with 3.5.1.12 at no, half, and full load at minimum, nominal, and maximum normal input voltages (see 3.5.1.1a).

4.5.12 Input power susceptibility (static inverters only).

4.5.12.1 Transient spike susceptibility. The inverter shall be operated at no load and full load at 28 VDC input. Spike voltages with repetition rates of 1, 10, 25, and 50 pulses per second shall be applied to the inverter input for both positive and negative polarities. The test duration shall be one minute for each combination of loading, pulse repetition rate, and spike polarity. The spike generator shall have a source impedance of 50 ±5 ohms, and shall produce a waveform as shown on figure 5 when operating into an open circuit prior to connection to the inverter input. The spike waveform at the inverter terminals during the test shall be recorded to aid in the verification of compliance. During the test, the inverter shall be observed for evidence of flashover or

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insulation breakdown, and the output shall be monitored to verify compliance with 3.5.1.13. Subsequent to the test, the output voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c.

4.5.12.2 Transient surge susceptibility. With the inverter operating at no load and full load with an input of 28 VDC, a positive step voltage of 80 volts with a duration of 50 milliseconds shall be applied to the inverter input. The transient surge shall be obtained from a low impedance source, such as a capacitor bank. The transient surge shall be applied four consecutive times for each load condition. The magnitude and duration shall be monitored on an oscilloscope and shall be measured, prior to application to the inverter under test, by applying and adjusting the transient across a dummy load which approximates the input of the inverter under test. The inverter shall be observed for evidence of flashover or insulation breakdown, and the output shall be monitored to verify compliance with 3.5.1.13. The inverter output voltage shall be permitted to drop to any level during the duration of the surge. Subsequent to the test the output voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c.

4.5.12.3 Ripple voltage susceptibility. With the inverter operating at no load, 1/4, 1/2, 3/4, and full rated load, unity power factor, at minimum, nominal, and maximum normal input voltage (see 3.5.1.1a), an audio frequency signal shall be superimposed on the inverter input and shall be varied from 10 hertz to 30 khertz (kHz). The amplitude of the signal shall conform to figure 6. At the option of the qualifying activity the inverter may be operated from any power source whose individual frequency characteristics conform to figure 6, and whose total peak-to-mean ripple is 2.0 volts or less. The output of the inverter shall be monitored throughout the test in order to measure output voltage, frequency, modulation, and waveform to verify compliance with 3.5.1.2, 3.5.1.3, 3.5.1.10, and 3.5.1.11.1. The test shall then be repeated at minimum rated lagging power factor load.

4.5.13 Extreme input voltage (static inverters only). With the inverter operating at no load, the input shall be varied between zero and 32 volts over a five-minute period. This test shall be repeated with the inverter operating at rated full load, unity power factor. The inverter shall then be operated for 1/2 hour at no load at each of the following input voltages: 32, 15, 10, and 5 VDC. This test shall be repeated with the inverter operating at rated full load, unity power factor. The output shall be monitored throughout the test to verify compliance with 3.5.1.14.

4.5.14 Reduced input voltage operation (rotary inverters only). The inverter shall demonstrate its ability to meet the requirements of 3.5.1.15 using the following method. Inverters rated 750 VA and below, with the exception of 10 VA inverters, shall demonstrate their ability to start and to operate continuously under full load at input voltages from 20 to 29 volts DC. Inverters rated above 750 VA shall start and operate with no load for at least 5 seconds with 22 volts DC input voltage.

4.5.15 Acoustical noise. Compliance to 3.5.1.16 shall be determined as follows: The acoustical noise of the inverter shall be measured in planes one foot from each of the six sides. In each plane, measurements shall be made opposite the center of the inverter surface to determine the sound level. The maximum sound level at all points shall be measured using the A-weighted scale of an exponentially averaging sound level meter. Measurements shall be made under anechoic

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conditions with the inverter supported in a manner to minimize acoustic transmission and reflection by the support structure.

4.5.16 Overspeed (rotary inverters only). The inverter shall be operated at 25 percent above rated speed at no load for five minutes without mechanical failure. Immediately following this test, the inverter shall pass the tests specified in 4.5.2.1a, b, and c (see 3.5.1.17).

4.5.17 Acceleration. The inverter shall be tested in accordance with the acceleration test of MIL-STD-810 Method 513, Procedures I and II. The test level for all directions in Procedure I (Structural Test – Non Operating) shall be accomplished using an acceleration level of 13.5g. The g level shall be applied along three mutually perpendicular axes in two opposite directions along each axis. The test time duration in each direction shall be at least one minute following centrifuge stabilization. At the conclusion of the test, the inverter shall be examined for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c. For Procedure II (Operational Test), the required acceleration level for all directions shall be 9.0 g, and shall be applied along three mutually perpendicular axes in two opposite directions along each axis. The test time duration in each direction shall be at least one minute following centrifuge stabilization and the item shall be operated continuously at 28 VDC input and full load during the test. The output voltage and frequency shall be monitored during the test to determine compliance with 3.5.2.2. At the conclusion of the test, the inverter shall be examined for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c.

4.5.18 Humidity. The inverter shall be tested in accordance with the humidity test of MIL-STD-810 Method 507 to determine compliance with 3.5.2.3. The inverter shall be subjected to a 24-hour conditioning period, followed by five cycles of the 48-hour aggravated temperature-humidity cycle. The inverter shall be operated with no load and nominal input voltage and frequency during the final 15 minutes of each 48-hour cycle. Following the test, the voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c. Evidence of corrosion, peeling, pitting, blistering, or cracking of the finish shall be cause for rejection.

4.5.19 Shock. The inverter shall be subjected to functional shock (Procedure I) and crash hazard shock (Procedure V) tests in accordance with Method 516 of MIL-STD-810 to determine compliance with 3.5.2.4. Calibration shocks shall be applied in accordance with Method 516 of MIL-STD-810 prior to shock testing.

4.5.19.1 Functional. The inverter shall be operating during the functional shock test at full load and 28 VDC input. The output voltage shall be monitored during the test to determine compliance with 3.5.2.4.1. The applied shock response spectrum (SRS) shall conform to the Functional Test for Flight Equipment spectrum of figure 516.6-8 of MIL-STD-810. The inverter shall be subjected to three shocks in each direction along three mutually perpendicular axes of the test item for a total of 18 shocks at the functional level. At the conclusion of the functional shock test, the inverter shall be inspected for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c.

4.5.19.2 Crash hazard. The inverter shall not be operated during the crash hazard shock test. The applied SRS shall conform to the Crash Hazard Test for Flight Equipment spectrum of

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figure 516.6-8 of MIL-STD-810. The inverter shall be subjected to two shocks in each direction along three mutually perpendicular axes of the test item for a total of 12 shocks at the crash hazard level. At the conclusion of the crash hazard shock test, there shall be no failure of the mounting attachment, the test item shall remain in place, and no parts shall have separated from the inverter. However, bending and distortion of the inverter shall be permitted.

4.5.20 Vibration. The inverter shall withstand the vibration environment induced by aircraft operations, as simulated by the vibration tests herein. Vibration testing shall be performed according to the vibration guidelines of MIL-STD-810, Method 514. After vibration exposure, component electrical performance shall be within specified limits and all component parts, connections and mounting hardware shall exhibit no evidence of cracking, rupture, or operational degradation. The inverter shall meet the requirements of 3.5.2.5.

4.5.20.1 Test fixture inspection and preparation. A vibration fixture survey shall be performed in accordance with test Method 514 of MIL-STD-810 using a dynamically equivalent test specimen. If a test specimen is not available, the actual inverter shall be used for the fixture survey (see 6.8.1). A sine sweep shall be performed: (1) after the fixture survey; (2) prior to each test excitation in the x, y, or z-axis; and (3) after each test excitation in the x, y, or z-axis. Examine each sine sweep to verify that no changes have occurred during the vibration testing in each orthogonal axis.

4.5.20.2 Inverters weighing less than 35 lb. The inverter shall be subjected to sinusoidal vibration applied to the mounting base. The amplitude shall be 0.015 inch (maximum total excursion 0.03 inch) up to 60 hertz. Above 60 hertz the acceleration shall be 10g. The inverter shall be vibrated for one excursion over the frequency range of 20 to 2000 hertz along each of its three mutually perpendicular axes. The vibration excursion shall take 15 minutes for each axis. Dwell time at resonance (20 to 2000 hertz) shall be 3 minutes at the worst resonance condition detected in each axis. During the test, the inverter shall be operated at full rated voltage and 28 VDC input. The output voltage modulation shall not exceed two percent, and output voltage and frequency shall not be outside the limits of 3.5.1.2 and 3.5.1.3 at any time during the test. Following the test the inverter shall pass the test specified in 4.5.2.1a, b, and c, and shall be thoroughly inspected for cracking or rupture of any components, solder connections, or of the mounting assembly itself. The presence of such damage shall be cause for rejection.

4.5.20.3 Inverters weighing 35 lb or more. The inverter shall be subjected to sinusoidal vibration applied to its mounting base along each of three mutually perpendicular axes of the inverter in accordance with figure 7. The total amount of test time per axis shall be three hours, with sinusoidal sweeps of 5-200-5 hertz and a sweep time of 36 minutes. The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 8. The sweep time of 36 minutes is that of an ascending plus a descending sweep and is twice the ascending sweep time shown on figure 8. The inverter shall be mounted without vibration isolators. During the test, the inverter shall be operated at full rated load and 28 VDC input. The output voltage modulation shall not exceed two percent, and output voltage and frequency shall not be outside the limits of 3.5.1.2 and 3.5.1.3 at any time during the test. Following the test, the inverter shall pass the test specified in 4.5.2.1a, b, and c, and shall be

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thoroughly inspected for cracking or rupture of any components, solder connections, or of the mounting assembly itself. The presence of such damage shall be cause for rejection.

4.5.21 Fungus. The inverter shall be tested in accordance with the fungus test of Method 508 of MIL-STD-810 to determine compliance with 3.5.2.6. At the conclusion of the test the inverter shall be removed from the test chamber, and excess moisture may be removed by turning the test item upside down or by shaking. No washing or wiping of the test item is permitted. The inverter shall be inspected for fungus growth, and the output voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c. Fungus testing is not required if all parts and materials are certified as non-nutrient supporting. A certificate signed by an individual authorized to speak for the manufacturer may be submitted in lieu of fungus testing with the approval of the qualifying activity. The certificate shall state that the inverter components are constructed of non-nutrient materials and do not use any organic material.

4.5.22 Salt fog. The inverter shall be tested in accordance with the salt fog test of Method 509 of MIL-STD-810 to determine compliance with 3.5.2.7. The inverter shall not be operating during exposure. At the completion of the exposure period, the inverter shall be dried by allowing the inverter to stand at room ambient conditions for 24 hours. The inverter shall be inspected both inside and out. Metal parts shall be examined for corrosion, moving parts for clogging or binding, and paint for blistering. Output voltage and frequency regulation shall then be measured as specified in 4.5.2.1a, b, and c. This regulation test shall be repeated after 48 hours additional exposure at room ambient conditions.

4.5.23 Dust. The inverter shall be tested in accordance with the blowing dust test (Procedure I) of Method 510 of MIL-STD-810 and as outlined below (see 3.5.2.8).

a. Dust composition shall be silica flour.

b. For step 3 of Procedure I, the inverter shall be operated at no load with an input voltage of 28 VDC for six hours at 23 °C (dust concentration of 0.3 ±0.2 gram per cubic foot and air velocity of 1,750 ±250 feet per minute). The chamber temperature shall be increased to 63 ºC during step 4. For step 7, the inverter shall be operated at no load for 6 hours at 63 ºC (same blowing dust conditions as step 3).

c. At the conclusion of the test the inverter shall be removed from the chamber and allowed to cool to room temperature. Accumulated dust shall be removed from the inverter by brushing, wiping, or shaking, with care taken not to introduce additional dust into the specimen. Under no circumstances shall either blast or vacuum cleaning be used to remove dust. The inverter shall be tested for normal operation, and its output voltage and frequency regulation shall be measured as specified in 4.5.2.1a, b, and c.

4.5.24 Explosive atmosphere. Determine conformance to the requirements of 3.5.2.9. The inverter shall be tested in accordance with Method 511 of MIL-STD-810.

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4.5.25 Effects of fluids. Determine conformance to the requirements of 3.3.4 by test or by similarity (see 6.2). Effects of fluids testing shall be conducted in accordance with Method 504 of MIL-STD-810.

4.5.26 Toxicity and fire resistance. Determine conformance of inverter materials to the requirements of 3.3.5 by test or by similarity (see 6.2). If testing is required, the inverter shall be subjected to a flammability test in accordance with Method 111 of MIL-STD-202.

4.5.27 Electrostatic discharge. The electrostatic discharge requirement of 3.5.2.10 shall be verified by the electrostatic discharge test of section 25 of DO-160. Test points shall be the top of the inverter case, the outside of the J1 connector, and the outside of the J2 connector.

4.6 Reliability development. The reliability development test shall consist of the tests specified in 4.6.1 and 4.6.2. The test shall be structured as a test, analyze and fix (TAAF) program wherein any failure (any operation outside of specified limits) which occurs is analyzed and corrective action implemented by the manufacturer to prevent recurrence of the failure. A minimum of 4000 hours total inverter operating time on two samples is required with two or less failures in order to successfully complete the test. In the event more than two failures occur during the 4000-hour period, additional inverter operation under the conditions of 4.6.2 is required until the total inverter operating hours divided by the number of failures (cumulative mean time between failures) equals 2000 or higher. When a corrective action has resolved a failure to the satisfaction of the qualifying activity all but the first occurrence of the resolved failure shall be dropped from consideration. No single inverter of the two samples shall accumulate less than 25 percent of the total test time, and both inverters shall complete the entire test of 4.6.1.

4.6.1 Temperature cycling. Two sample inverters shall be operated for 50 hours under the input voltage, load, and duty cycle profile shown in figure 9. Twenty-five of the operating hours shall be at unity power factor, and the remaining 25 hours at the minimum lagging power factor shown on the applicable specification sheet. During the test the ambient temperature shall be varied continuously between -55 °C and +71 °C with the extreme temperatures maintained for 1 ±0.25 hours prior to ramping to the opposite extreme temperature. The heating and cooling ramps shall be performed at a minimum rate of 3 °C/minute. This test may be performed continuously or in increments equal to one complete temperature cycle. During the test, the output voltage and frequency shall be monitored for compliance with 3.5.1.2 and 3.5.1.3. Following the test, the inverter shall pass the test of 4.5.2.1.

4.6.2 Room ambient operation. The two sample inverters shall be operated at room ambient conditions under the input voltage, load, and duty cycle profile shown in figure 9 for the time duration required by 4.6. During the test, the output voltage and frequency shall be monitored for compliance with 3.5.1.2 and 3.5.1.3. Each inverter shall be subjected to the test of 4.5.2.1 after each 250 operating hours. Following the entire test duration, each inverter shall pass the following tests at room temperature only: 4.5.2.1, 4.5.2.2, 4.5.3, 4.5.7, 4.5.9, 4.5.10.1.1a and b.

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4.7 Disassembly and inspection. At the conclusion of the test program, each sample shall be disassembled, as necessary, to inspect for evidence of wear, defects, and overheating. The presence of such evidence shall be cause for rejection.

4.8 Burn-in (static inverters only). The inverter shall be subjected to one or more cycles of testing conforming to figure 10. The number of cycles shall be determined by referencing table VII. The heating and cooling temperature ramps on figure 10 shall be performed at a minimum of 5 °C/minute. The inverter shall be turned off during the cool down period (cooling ramp) and turned on for the balance of each cycle. The equipment shall be vibrated in one axis with amplitude of $0.02g^2$/Hz over a frequency range of 20-2,000 Hz during the last 20 minutes of each four-hour stabilized temperature condition. Failures may occur and be corrected in any cycle prior to the final cycle. The burn-in is considered acceptable if there is no failure during the final cycle. If a failure occurs in the final cycle, the equipment failure may be corrected, subject to approval by the Government and a failure-free cycle run.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items described herein are intended for use on aircraft to convert 28 Volt DC power to either single- or three-phase, 115 VRMS, 400 hertz AC power. Inverters typically are used to provide AC power on aircraft with main electrical generating systems that provide DC power. The military operating environment and electrical performance requirements of this specification exceed the requirements used for commercial aircraft inverters.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification and applicable specification sheet.

b. Class of inverter.

c. Military part number.

d. Packaging requirements (see 5.1 and 6.4).

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e. Reinspection date marking and any special marking (see 3.7.2).

f. Handbooks or manuals, when required.

g. Contract provisions for acquisition of manufacturer's data (see 4.2.1) may include:

(1) Schedules for component design, development, and delivery;
(2) System and component wiring diagrams, schematics, outline drawings;
(3) List of parts, list of materials, functional descriptions, operating instructions, installation instructions, and servicing requirements (see 3.3, 3.4);
(4) Contractor analyses of sneak circuits, failure modes and effects, parts stress, parts failure rate modeling, and reliability prediction (see 3.6);
(5) Stress screening test procedures, acceptance test procedures, manufacturer's test data, and qualification by similarity data (see 4.5.21, 4.5.25, and 4.5.26);
(6) Verification data, manufacturing process controls, inspection criteria, configuration management data, engineering change processes, and parts traceability (see 3.8).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-7032 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command (AIR-4.4.5.2), 48298 Shaw Road, Patuxent River, MD 20670-1900. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.daps.dla.mil.

6.3.1 Provisions governing qualification. Qualification will be conducted in accordance with the provisions and guidelines of SD-6. Copies of this document are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.3.2 Aircraft demonstration (ground and flight). Prior to flight, the inverter will be subjected to an aircraft ground demonstration for safety of flight. Inverter conformance to specification requirements will be demonstrated throughout all aircraft operations, and the cumulative operating flight time should be not less than 50 hours.

6.4 Packaging. Preservation and packaging should be in accordance with level A or C of MIL-STD-2073-1, as specified by the contract (see 5.1 and 6.2.d).

6.5 Definitions. The definitions in MIL-STD-704, MIL-STD-810, and MIL-STD-461 will apply in addition to those cited herein.

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6.5.1 Abnormal input voltage range. The abnormal input voltage range is the voltage range encountered during operation from aircraft batteries or other emergency power sources.

6.5.2 Alternating voltage and current. Alternating voltages and currents used in this specification are root mean square (RMS) values, unless otherwise indicated.

6.5.3 Crest factor. The crest factor is the absolute value of the ratio of the peak to the RMS value for each half cycle of the voltage waveform.

6.5.4 Direct voltage and current. Direct voltages and currents used in this specification are mean (average) values.

6.5.5 Full-rated load (full load). Full-rated load used in this specification is the maximum continuous inverter output load specified in the applicable specification sheet for a given temperature-altitude condition.

6.5.6 Load. Single-phase load is considered the product of the RMS line current and the RMS voltage measured at the terminals of the inverter. Three-phase load is the summation of the three individual loads (phase voltage times phase current) measured at the terminals of the inverter.

6.5.7 Low input voltage range. The low input voltage range is the voltage range encountered during operation from aircraft batteries when those batteries are being used for engine starting.

6.5.8 Normal input voltage range. The normal input voltage range is the voltage range encountered when all aircraft engines are operating and the aircraft electrical system is fully operational.

6.5.9 Output voltage unbalance. Output voltage unbalance is the maximum difference between phase voltage magnitudes.

6.5.10 Phase voltage. Phase voltage is the line-to-neutral for wye connection and line-to-line for delta connection.

6.6 Standard design practices. The following paragraphs contain design guidelines based on actual experience that should be considered by the manufacturer when an inverter is to be developed to meet the performance requirements of this specification.

6.6.1 Quality assurance and configuration management. As required by the contract, quality assurance management and failure analysis/reporting should be in accordance with industry standards, such as ISO 9000. Product identification, parts traceability, product configuration control and documentation data control should be in accordance with industry standards for configuration management, such as TechAmerica/ANSI EIA-649 or ISO 10007. MIL-HDBK-61 provides guidance on configuration management and TechAmerica/ANSI EIA-649 configuration control process. Copies of ISO 9000 and ISO 10007 are available from info@ansi.org or American National Standards Institute, 11 West 42nd Street, 13th Floor, New

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York, New York, 10036. Copies of TechAmerica/ANSI EIA-649 are available from TechAmerica, 601 Pennsylvania Ave., NW, North Building, Suite 600, Washington, DC 20004. Copies of MIL-HDBK-61 are available from https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.6.2 Materials and structural design. Military standard parts are desired. Commercial industry parts, such as screws, bolts, nuts and washers, may be used provided that performance requirements are met under all specified operating conditions and environments. Design characteristics and limitations should be observed for the materials and components selected for use in the inverter.

6.6.2.1 Corrosion. Corrosion resistance treatments for aluminum alloy materials are provided in MIL-A-8625. Corrosion protection methods for magnesium alloy materials are provided in SAE-AMS-M-3171.

6.6.2.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. Information regarding toxic chemicals, hazardous substances, and ozone depleting substances can be found at the Environmental Protection Agency web site at http://www.epa.gov/.

6.6.2.3 Fasteners, inserts and retention. Screws, fasteners, inserts should be selected using industry standard NASM 1515 for guidance. Stud nuts should be self-locking steel nuts with captive washer in accordance with NASM 90415, or equivalent.

6.6.2.4 Bearing installation and lubrication. Steel sleeve bearing inserts are recommended when bearing support structure is made of non-ferrous material. The outer ring for each shaft bearing should be retained axially and should have a locking method to prevent rotation. Bearing inner rings should rotate with the drive shaft. The locking methods for the inner and outer bearing rings should allow for thermal expansion differentials. The grease type selected for bearing lubrication should be capable of a minimum of 100 hours, and withstand continuous operation at bearing temperatures from -40 °C to +205 °C.

6.6.2.5 Cooling fan. Inverters for which an internally mounted, integral cooling fan is permitted (see 3.4.3.5) should use a motor having ball bearings of quality level ABEC-3 or higher, and the fan should be rated for operation at 100 °C ambient or higher.

6.6.3 Electrical and electronics design.

6.6.3.1 Wiring and distribution. External electrical wiring for power distribution, component interconnection and system control should conform to SAE-AS22759. Load contactor relays should conform to MIL-PRF-6106 standards. Power feeder sizes should be selected using SAE-AS50881 guidelines for conductor current-carrying capacity.

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6.6.3.2 Electronics internal wiring and electric connections. Printed circuit board design and performance criteria are provided in MIL-PRF-31032, MIL-HDBK-1547, MIL-HDBK-1861, and industry standards IPC-2221, and IPC-D275. Guidelines 1 and 69 of MIL-HDBK-454 provide guidance on internal wiring practices. Copies of IPC 2221 and IPC-D275 are available from http://www.ipc.org/ or IPC - Association Connecting Electronics Industries, 3000 Lakeside Drive Suite 309 S, Bannockburn, Illinois, 60015-1249.

6.6.3.3 Semiconductors, resistors, and capacitors. MIL-STD-750 and MIL-PRF-19500 provide guidance on the design and test of semiconductor devices. Adjustable resistors should conform to MIL-PRF-39015. MIL-HDBK-198 provides guidance on capacitor selection. The use of wet slug tantalum capacitors is not recommended.

6.6.3.4 Microelectronic devices. Microelectronic devices should be selected in accordance with Guideline 64 of MIL-HDBK-454. Microelectronic devices qualified to MIL-STD-883, and listed in MIL-PRF-38535, are preferred. Static discharge control should be in accordance with MIL-STD-1686, or an equivalent industry standard (such as JEDEC JESD625).

6.6.3.5 Parts de-rating. DoD standard practices for electrical and electronic parts de-rating are documented in SD-18 which is available at the NSWC-Crane website http://www.navsea.navy.mil/nswc/crane/sd18/Pages/ContactUs.aspx. To obtain a CD-ROM copy, send your request to Commander, Naval Sea Systems Command – Crane Division, ATTN: Code 6024, Building 2940 West, 300 Highway 361, Crane, IN 47522.

6.6.4 Reliability. Reliability should be considered in each phase of the inverter design process. MIL-STD-790, MIL-HDBK-217, and MIL-HDBK-251 provide guidance on established reliability and reliability prediction.

6.6.5 Maintainability. The inverter should be designed to facilitate assembly, disassembly, location of trouble sources, and maintenance without the aid of special tools. All functional parts should be readily identifiable, accessible, and removable for replacement. When printed circuit boards are utilized, only clear transparent conformal coating should be used in order to permit identification, location, removal, and replacement of functional parts.

6.6.6 Failure mode and effects analysis (FMEA). A failure mode and effects analysis should be performed on the inverter. Internal protective functions should be independent of the control functions, and no single fault or failure should result in inverter operation outside the limits specified herein.

6.6.7 Identification markings. MIL-STD-130, MIL-STD-1285, and MIL-HDBK-1812 provide guidelines for identification marking of equipment and parts.

6.7 Installation. The aircraft manufacturer should allow clearance of at least one inch at each end of the inverter in addition to the maximum dimensions shown on the applicable specification sheet for installation and ventilation.

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6.8 Vibration. Inverters meeting the vibration requirements specified herein are suitable for use in jet, propeller and helicopter applications.

6.8.1 Vibration test fixture. Where impedance mismatch between platform/material and laboratory vibration exciter/test item are significantly different, force control or acceleration limiting control strategies may be required to avoid unrealistically severe vibration response. If a dynamically equivalent test specimen is not available for the vibration fixture survey, then the actual component may be utilized provided the evaluation level does not exceed 20 percent of the actual rms random specified value or 1g (peak) when running sine vibration.

6.9 Aviation fuel and hydraulic fluids. Testing for resistance to the effects of oil, fuel and hydraulic fluids should include those listed in MIL-PRF-23699, MIL-PRF-7808, DOD-PRF-85734 and MIL-DTL-5624 (JP-4/JP-5), MIL-PRF-8188, MIL-PRF-5606, and MIL-PRF-87257.

6.10 Three-phase application. For three-phase aircraft application, the load on any one phase should not exceed one third of the total three-phase inverter rating on a steady-state basis.

6.11 Subject term (key word) listing.

AC supply
Electrical
Single-phase
Static supply
Three-phase

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.13 Specification sheet requirements. Specification sheets define the performance, design, interface, test, qualification, etc. requirements for a specific inverter. The paragraphs and tables in this specification that are identified in table VIII, require information to be provided in the specification sheet. This information will provide the minimum requirements to define the specific inverter. The following applies: Paragraphs with an X in the “As Required By” column will either have a default value given in this specification or must have a value or requirement given in the specification sheet. Paragraphs with an X in the “Unless Otherwise Specified” column will have the requirement or value given only in the specification sheet, if it is required.

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TABLE I. Static inverter efficiency.

Rating (VA)	Minimum efficiency (percent)
100 and below	60
101 through 1000	65
1001+	70

TABLE II. Input ripple current.

Inverter rating (VA)	Peak-To-Mean input ripple current max (Amp)
Under 30	0.5
30-90	1
91-200	2
201-500	5
501-1000	10
1001-2500	20
Over 2500	25

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TABLE III. Qualification inspection.

Inspection title (examination or test)	Requirement paragraph	Test method paragraph	Qualification (sample no.)			
			1	2	3	4
Visual and mechanical inspection	3.3.1, 3.3.7, 3.4, 3.4.1, 3.4.3.8, 3.7.1, 3.7.2, 3.7.3, 3.7.5, 3.8	4.5.1	X	X	X	X
Voltage and frequency regulation	3.5.1.2, 3.5.1.3	4.5.2				
At room ambient temp-altitude conditions	3.5.1.2, 3.5.1.3	4.5.2.1	X 1/	X 1/	X	X
Power factors	Specification sheet	4.5.2.2	X 1/	X 1/		
At maximum temp-altitude conditions	3.5.2.1	4.5.2.3	X 1/	X 1/		
At minimum temp-altitude conditions	3.5.2.1	4.5.2.4	X 1/	X 1/		
Operating position		4.5.2.5	X	X		
Worst condition		4.5.2.6	X	X		
Efficiency	3.5.1.4	4.5.3	X 1/	X 1/	X	X
Three-phase output	3.5.1.5	4.5.4	X 1/	X 1/	X	X
Recovery time	3.5.1.6	4.5.5	X	X	X	X
Input power application	3.5.1.7	4.5.6		X		X
Overload & short circuit capacity	3.5.1.8	4.5.7	X 1/	X 1/	X	X
Electromagnetic compatibility	3.5.1.9	4.5.8	X		X	
Output modulation	3.5.1.10	4.5.9	X 1/	X 1/	X	X
Waveform	3.5.1.11	4.5.10	X 1/	X 1/	X	X
Input ripple current (static inverters only)	3.5.1.12	4.5.11	X	X		
Input power susceptibility (static inverters only)	3.5.1.13	4.5.12	X	X		
Extreme input voltage (static inverters only)	3.5.1.14	4.5.13	X	X		
Reduced input voltage operation (rotary inverters only)	3.5.1.15	4.5.14	X	X		
Acoustical noise	3.5.1.16	4.5.15	X			
Overspeed (rotary inverters only)	3.5.1.17	4.5.16			X	
Acceleration	3.5.2.2	4.5.17	X			
Humidity	3.5.2.3	4.5.18	X			
Shock	3.5.2.4	4.5.19	X			
Vibration	3.5.2.5	4.5.20		X	X	
Fungus	3.5.2.6	4.5.21				X
Salt fog	3.5.2.7	4.5.22		X		

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TABLE III. Qualification inspection – Continued.

Inspection title (examination or test)	Requirement Paragraph	Test method paragraph	Qualification (sample no.)			
			1	2	3	4
Dust	3.5.2.8	4.5.23				X
Explosive atmosphere	3.5.2.9	4.5.24			X	
Reliability development	3.6.1	4.6			X	X
Disassembly and inspection	---	4.7			X	X

NOTE:

1/ For three-phase inverters all tests shall be conducted in the three-phase wye configuration. For inverters designed to operate in both the wye and delta configuration, tests indicated by 1/ shall be conducted in both wye and delta configurations.

TABLE IV. Conformance inspection.

Inspection title (examination or test)	Requirement Paragraph	Test method paragraph
Visual and mechanical inspection	3.3.1, 3.3.7, 3.4, 3.4.1, 3.4.3.8, 3.7.1, 3.7.2, 3.7.3, 3.7.5, 3.8	4.5.1
Burn-in (static inverters only)	3.8.1	4.8
Overload and short circuit capacity (static inverters only)	3.5.1.8	4.5.7
Voltage and frequency regulation at room ambient temperature -altitude conditions	3.5.1.2, 3.5.1.3	4.5.2.1
Power factors	Specification sheet	4.5.2.2
Efficiency	3.5.1.4	4.5.3
Output voltage unbalance (three-phase inverters only)	3.5.1.5.2	4.5.4.1
Phase rotation (three-phase inverters only)	3.5.1.5.1	4.5.4.2
Input power application	3.5.1.7	4.5.6
Output voltage modulation	3.5.1.10.1	4.5.9.1

NOTE:

For three-phase inverters, all tests shall be conducted in the three-phase wye configuration.

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TABLE V. Maximum temperature-altitude conditions.

Class A inverters		Class B inverters		Class C inverters	
Ambient temperature (°C)	Altitude (Feet)	Ambient temperature (°C)	Altitude (Feet)	Ambient temperature (°C)	Altitude (Feet)
+71	Sea level	+85	Sea level	+120	Sea level
+25	15,000	+85	17,500	+80	25,000
+8	25,000	+50	35,000	+60	37,000
-10	35,000	+20	50,000	+40	50,000
-10	50,000	+20	1/ 65,000	-12	65,000

1/ Class B power rating lowered 50 percent above 50,000 feet.

TABLE VI. Minimum temperature-altitude conditions.

Classes A, B, and C inverters	
Ambient temperature (°C)	Altitude (Feet)
-55	Sea level
-55	10,000
-55	25,000
-55	50,000
-55	1/ 65,000

1/ Class B power rating lowered 50 percent above 50,000 feet.

TABLE VII. Number of burn-in cycles.

Number of electronic parts	Number of cycles
0 – 100	1
101 – 500	3
501 – 2000	6
2001+	10

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TABLE VIII. Specification sheet requirements.

Specification Paragraphs	As Required By	Unless Otherwise Specified
3.4.2.1.1	x	
3.4.2.1.1.3	x	
3.4.2.2.2.1	x	
3.4.3.5	x	
3.4.3.8	x	
3.5.1.1a	x	
3.5.1.1b	x	
3.5.1.1c	x	
3.5.1.3	x	
3.5.1.4	x	
3.5.1.7	x	
3.5.1.8.1	x	
3.5.1.16		x
3.5.2.1	x	
3.7.1	x	
3.7.4	x	
4.2.2	x	
4.2.3	x	
4.4.1		x
4.4.1.6	x	
4.5.1	x	
4.5.2.2	x	
4.5.10.1.1b	x	
4.6.1	x	
TABLE III	x	
TABLE IV	x	

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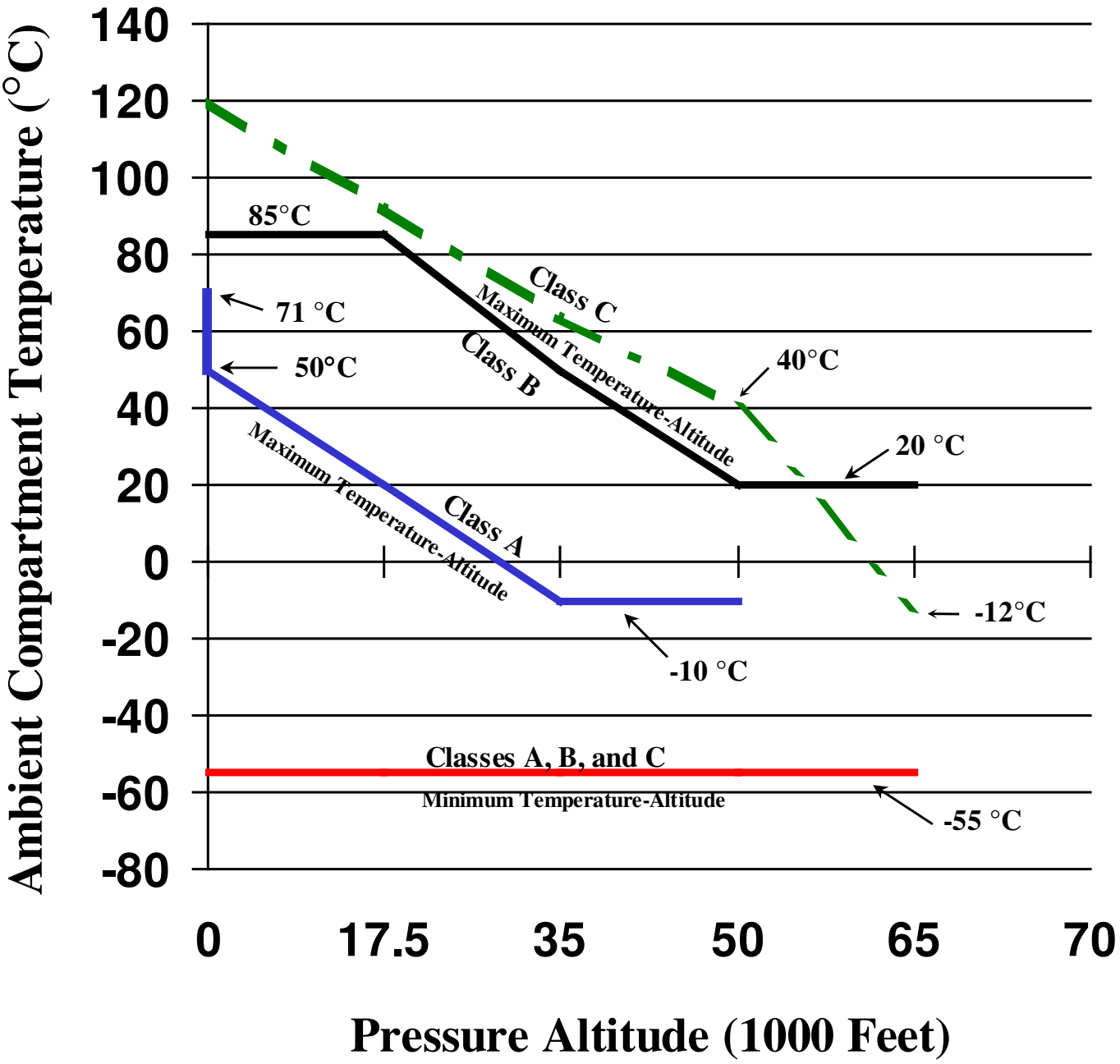


FIGURE 1. Temperature – altitude.

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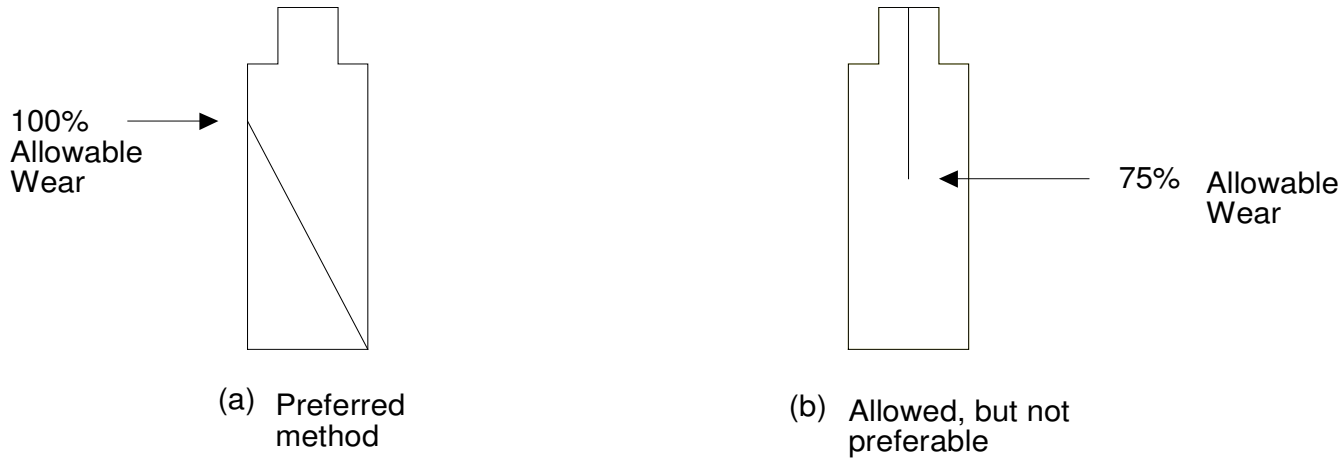


FIGURE 2. Brush markings.

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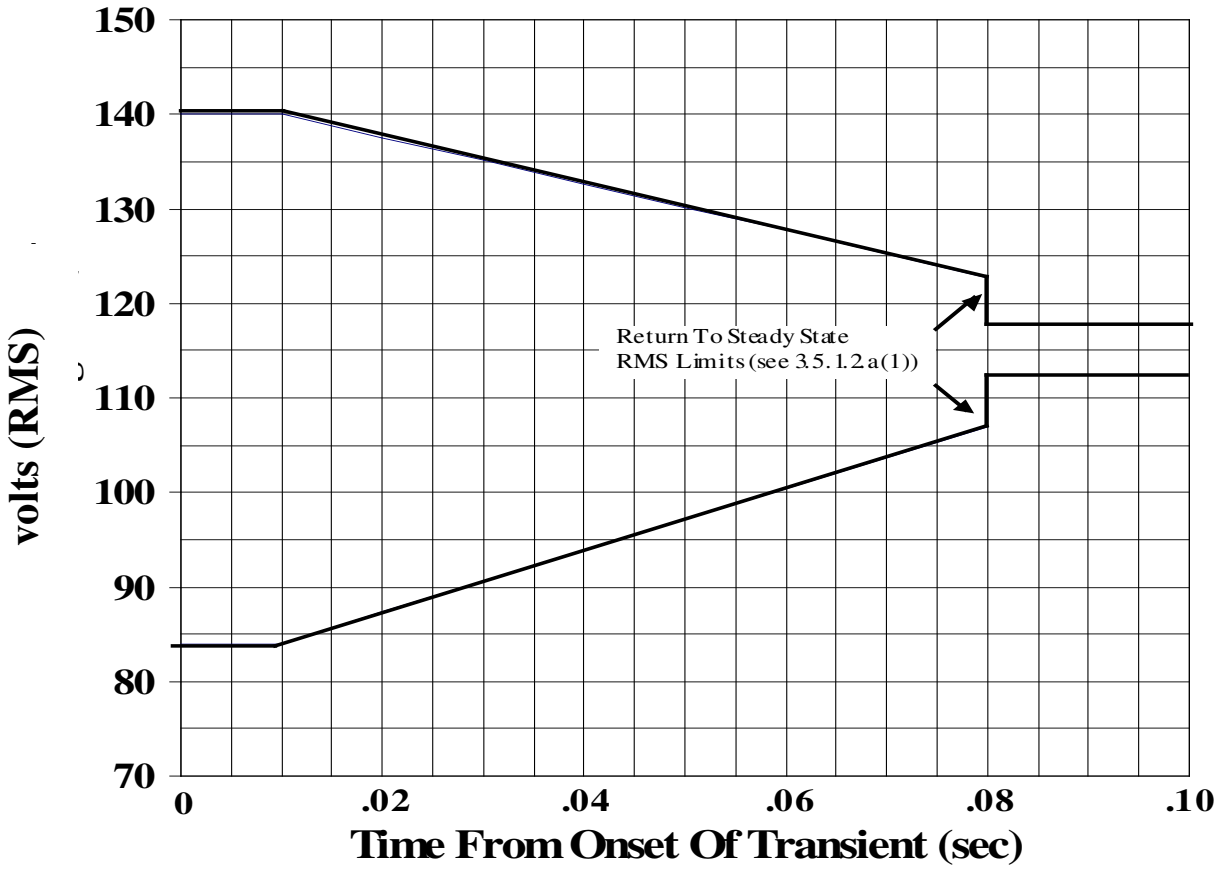


FIGURE 3. Inverter output voltage transient limits.

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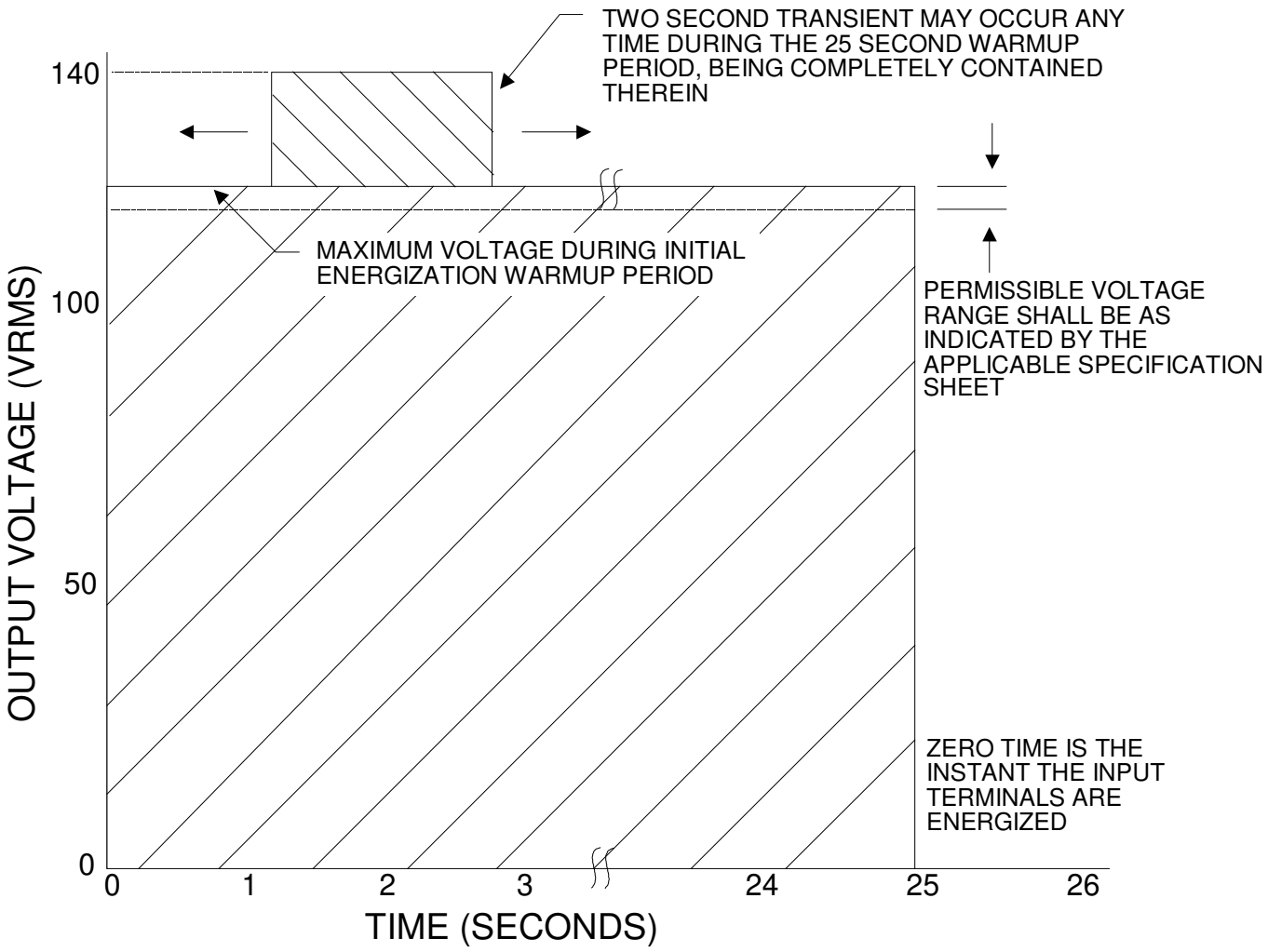


FIGURE 4. Output voltage during input power application.

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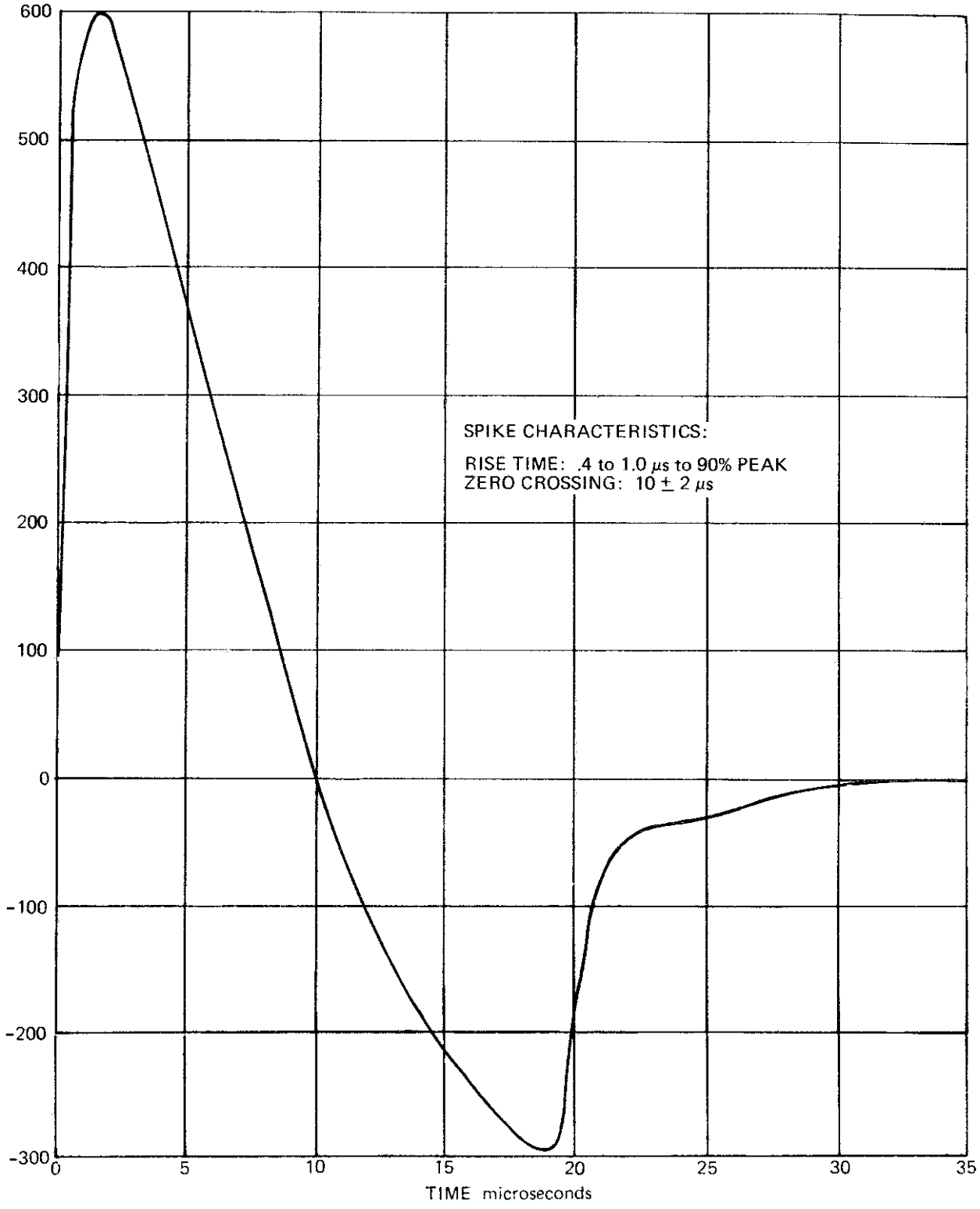


FIGURE 5. Spike susceptibility waveform.

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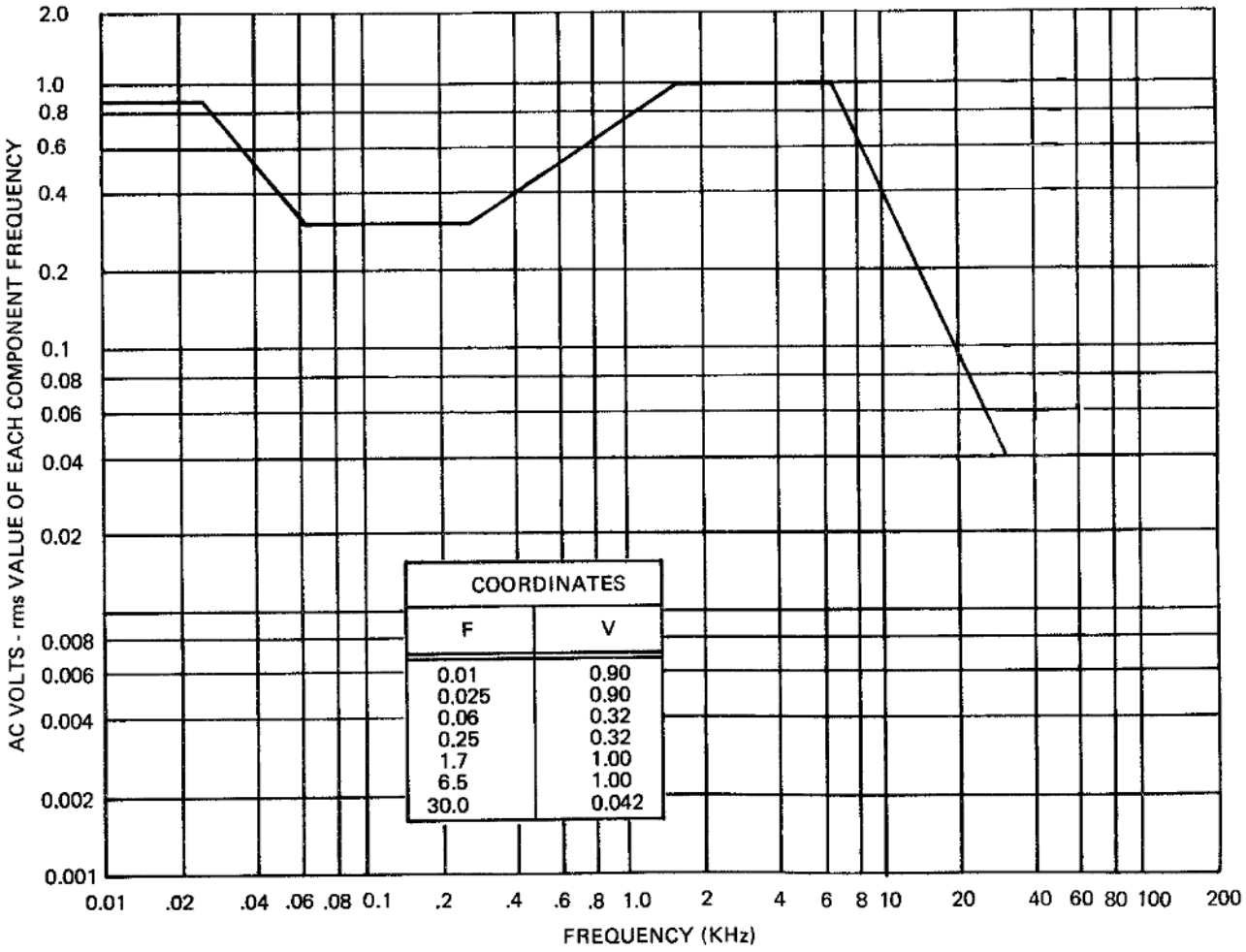


FIGURE 6. Frequency characteristics of ripple for ripple susceptibility test.

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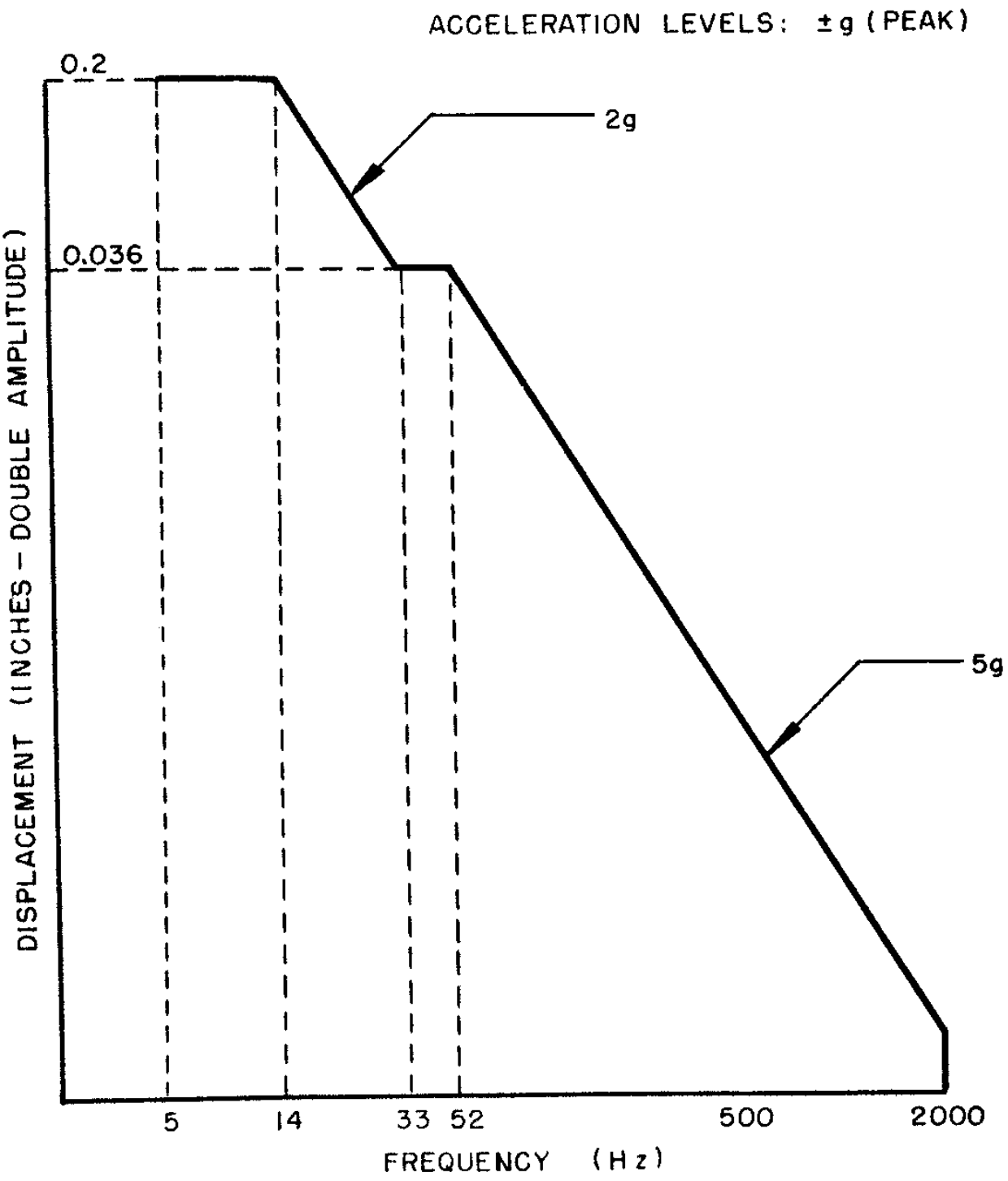


FIGURE 7. Vibration test curve.

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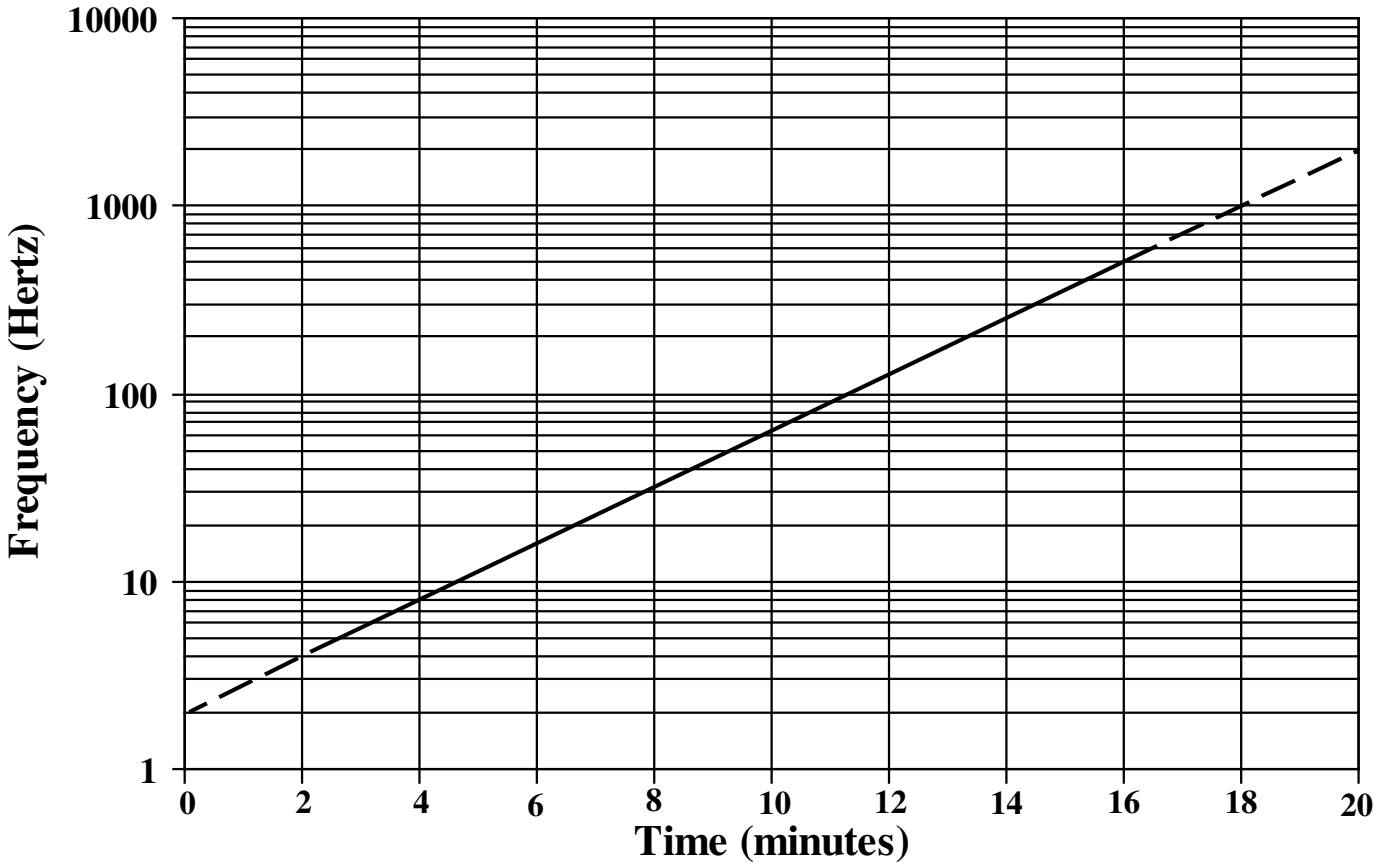


FIGURE 8. Logarithmic sweep.

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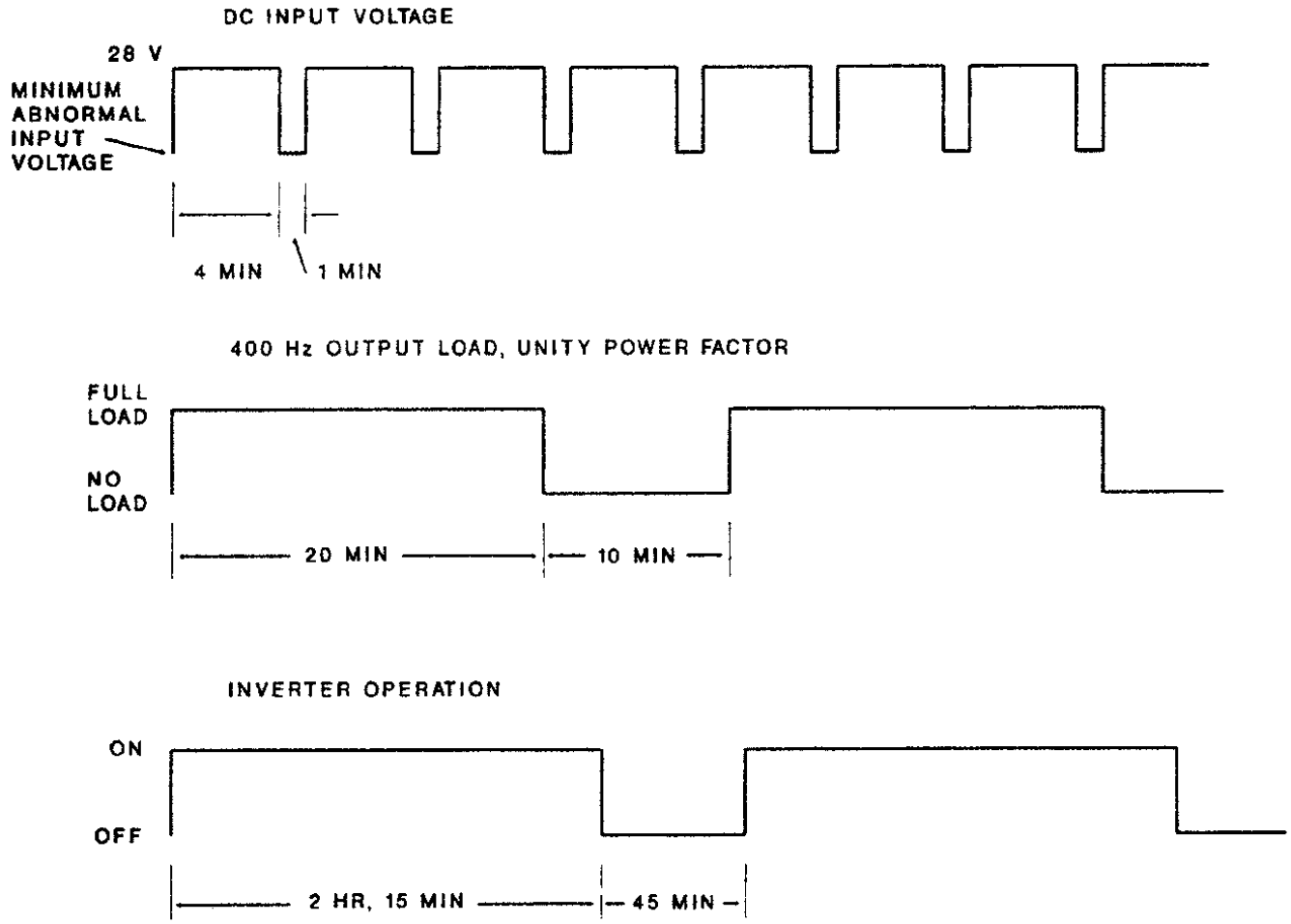


FIGURE 9. Input voltage, load, and duty cycle profile.

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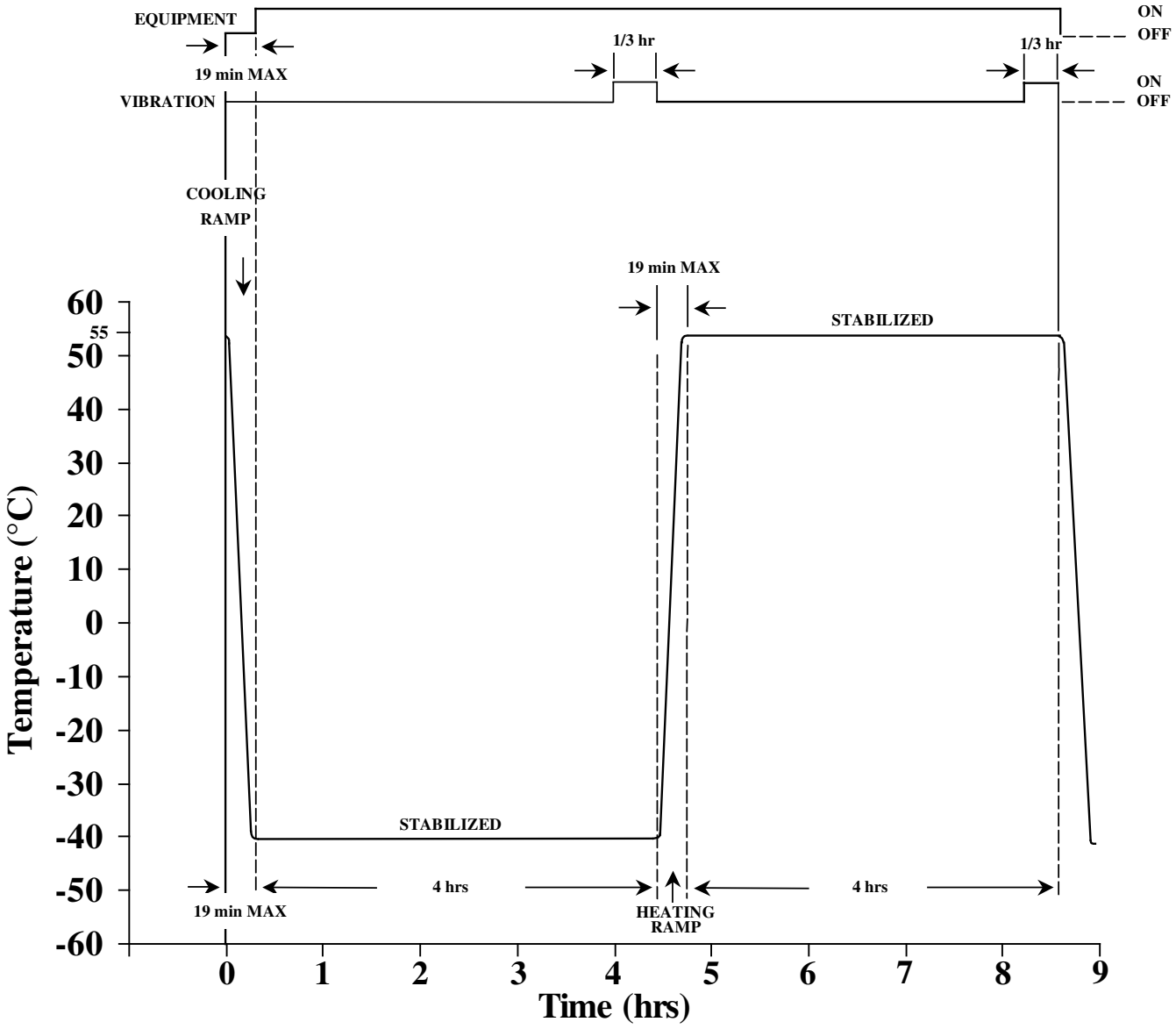


FIGURE 10. Typical burn-in test cycle.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – AS
Air Force – 99

Preparing activity
Navy - AS

(Project 6125-2008-001)

Review activities:
Army – CR4
Air Force – 71
DLA – GS

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